

Universal Linear Algebra Performance Audit (FAST) (FAST)

Audit Context:

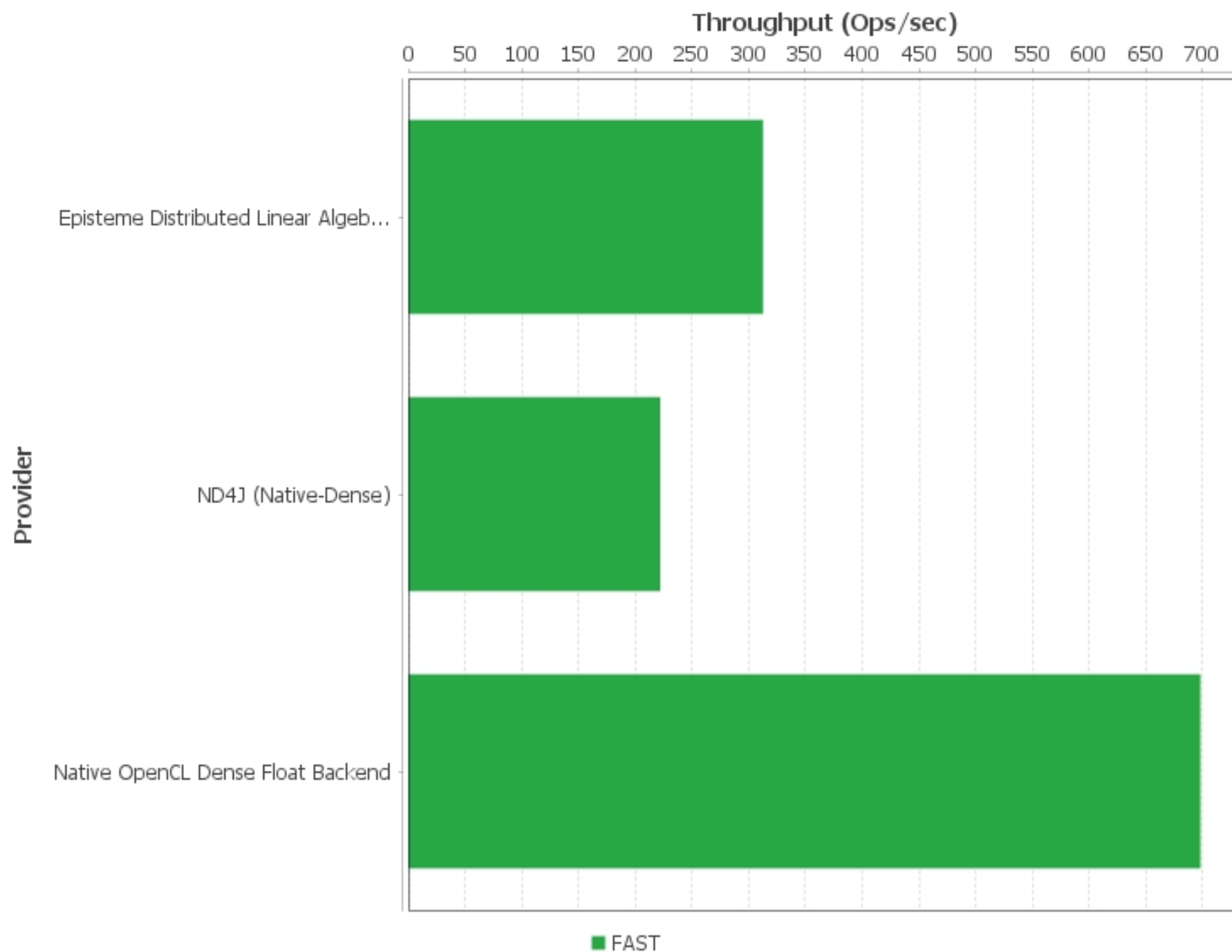
Precision: fast
Environment: GCP
Matrix Size: 5x5
Reconstructed: True (Live from JSON)

Provider	Operation	Domain	Result Summary
DistributedLinearAlgebraProvider	Episteme Distributed Linear Algebra Provider (LocalDistributedContext)	Linear Algebra Performance	39 operations (avg 7,980 ms)
NativeND4JLinearAlgebraBackend	ND4J (Native-Dense)	Linear Algebra Performance	17 operations (avg 3,847 ms)
NativeOpenCLDenseLinearAlgebraFloatBackend	Native OpenCL Dense Float Backend	Linear Algebra Performance	23 operations (avg 6,977 ms)
NativeOpenCLSparseLinearAlgebraFloatBackend	Native OpenCL Sparse Float Backend	Linear Algebra Performance	8 operations (avg 2,747 ms)
GRPCLinearAlgebraBackend	Episteme gRPC Remote (localhost:50051)	Linear Algebra Performance	No successful operations

Operation Comparison: R:Add (FAST)

Episteme Multimodal Audit | R:Add

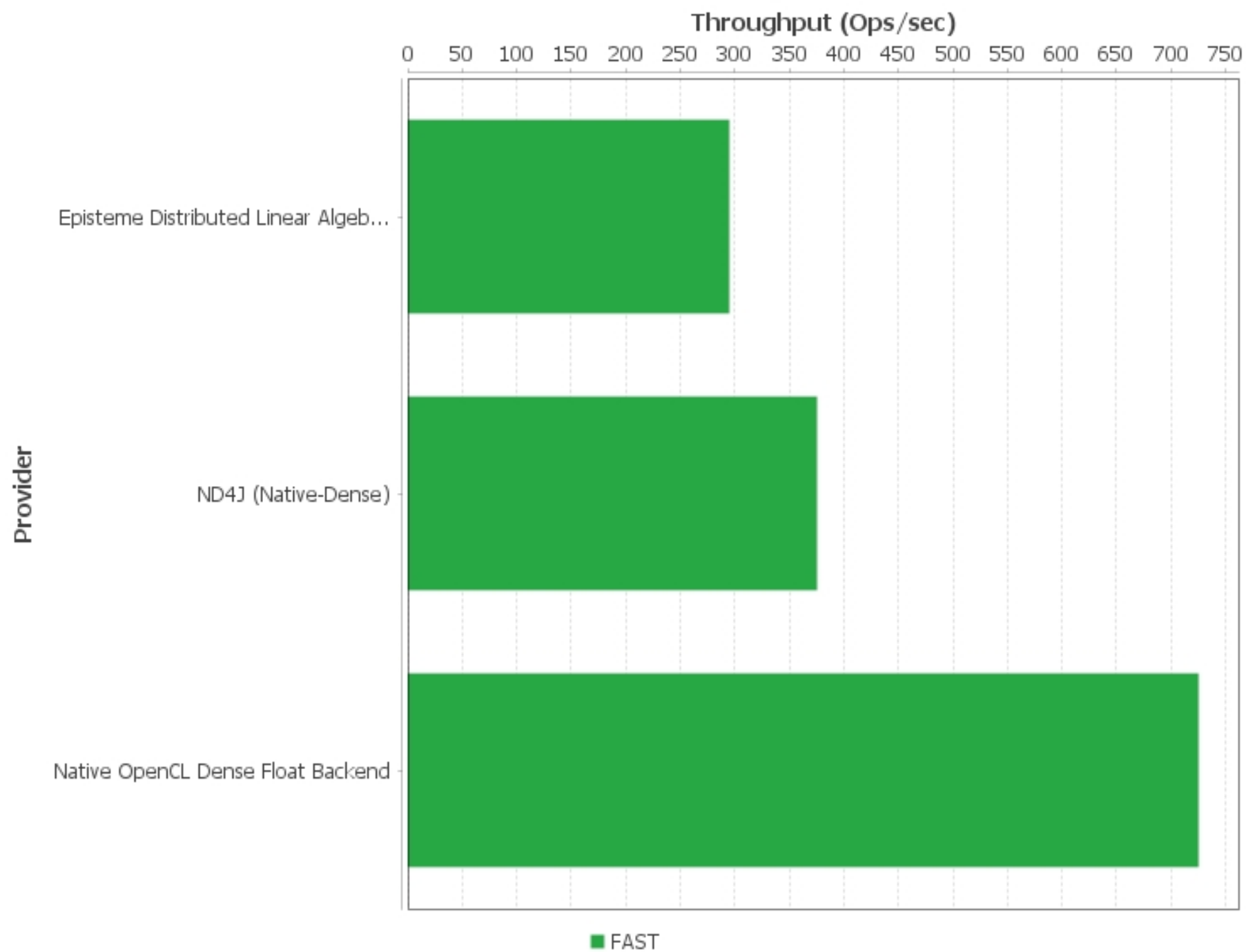
R:Add Performance Comparison



Operation Comparison: R:Sub (FAST)

Episteme Multimodal Audit | R:Sub

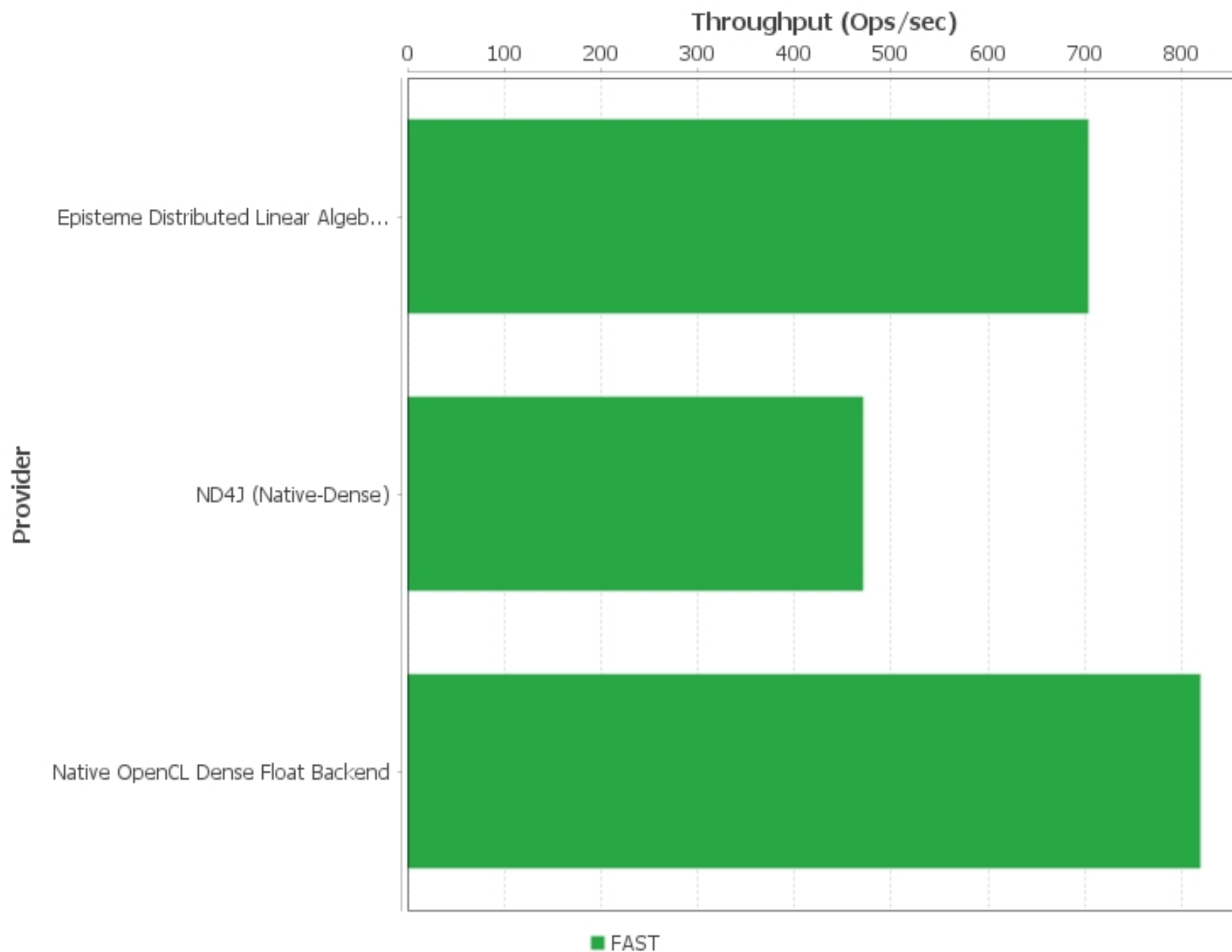
R:Sub Performance Comparison



Operation Comparison: R:Scale (FAST)

Episteme Multimodal Audit | R:Scale

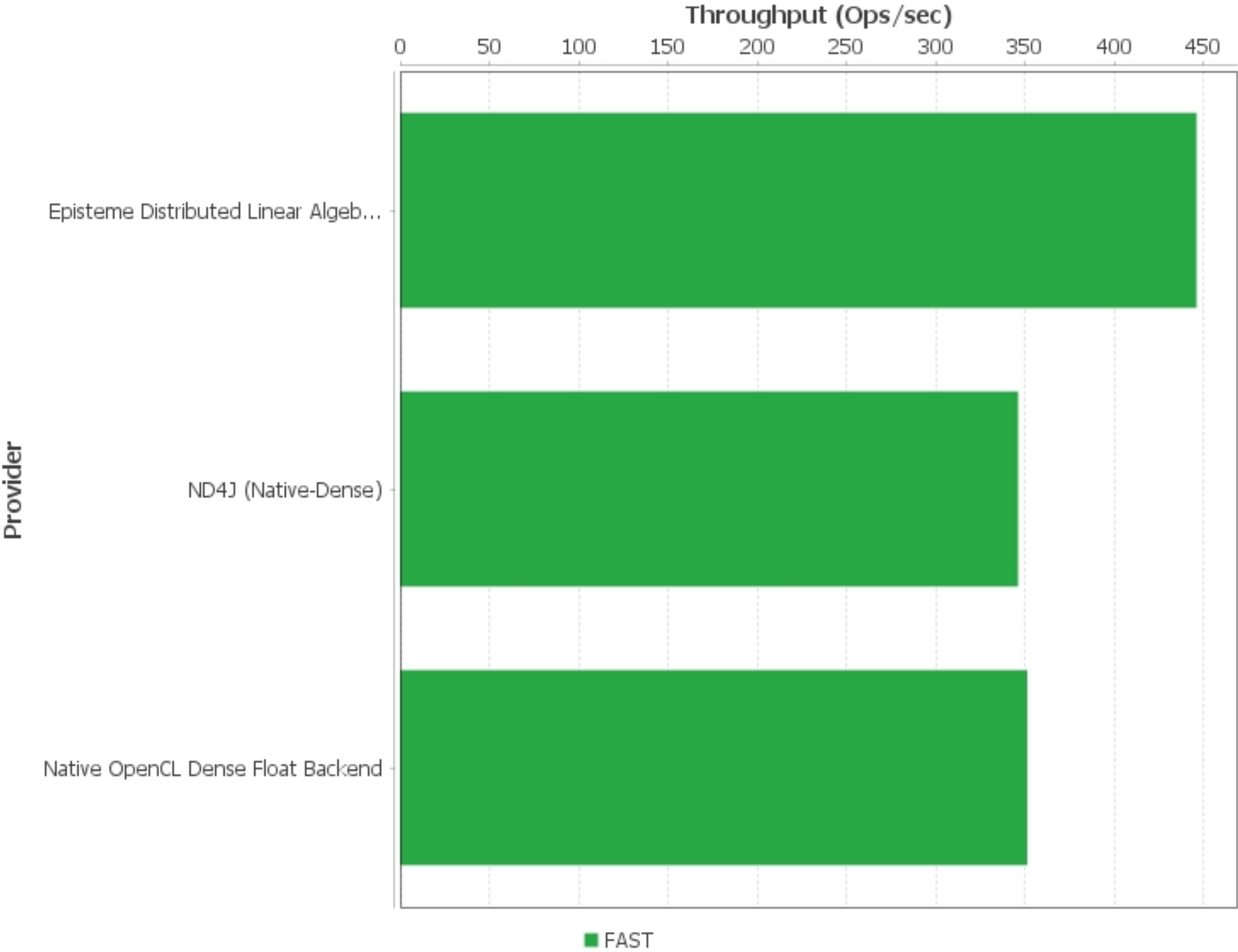
R:Scale Performance Comparison



Operation Comparison: R:Mul (FAST)

Episteme Multimodal Audit | R:Mul

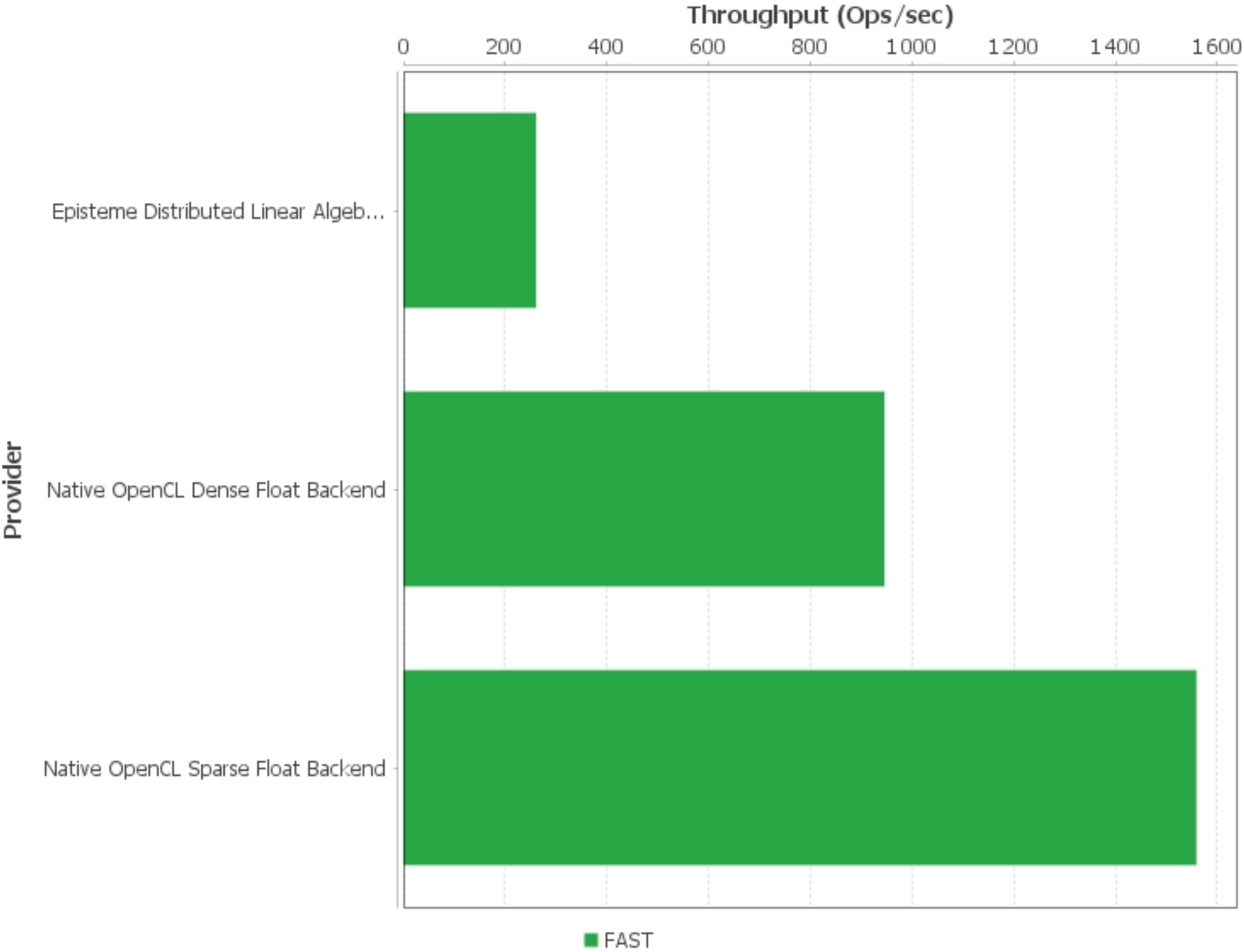
R:Mul Performance Comparison



Operation Comparison: R:MatVec (FAST)

Episteme Multimodal Audit | R:MatVec

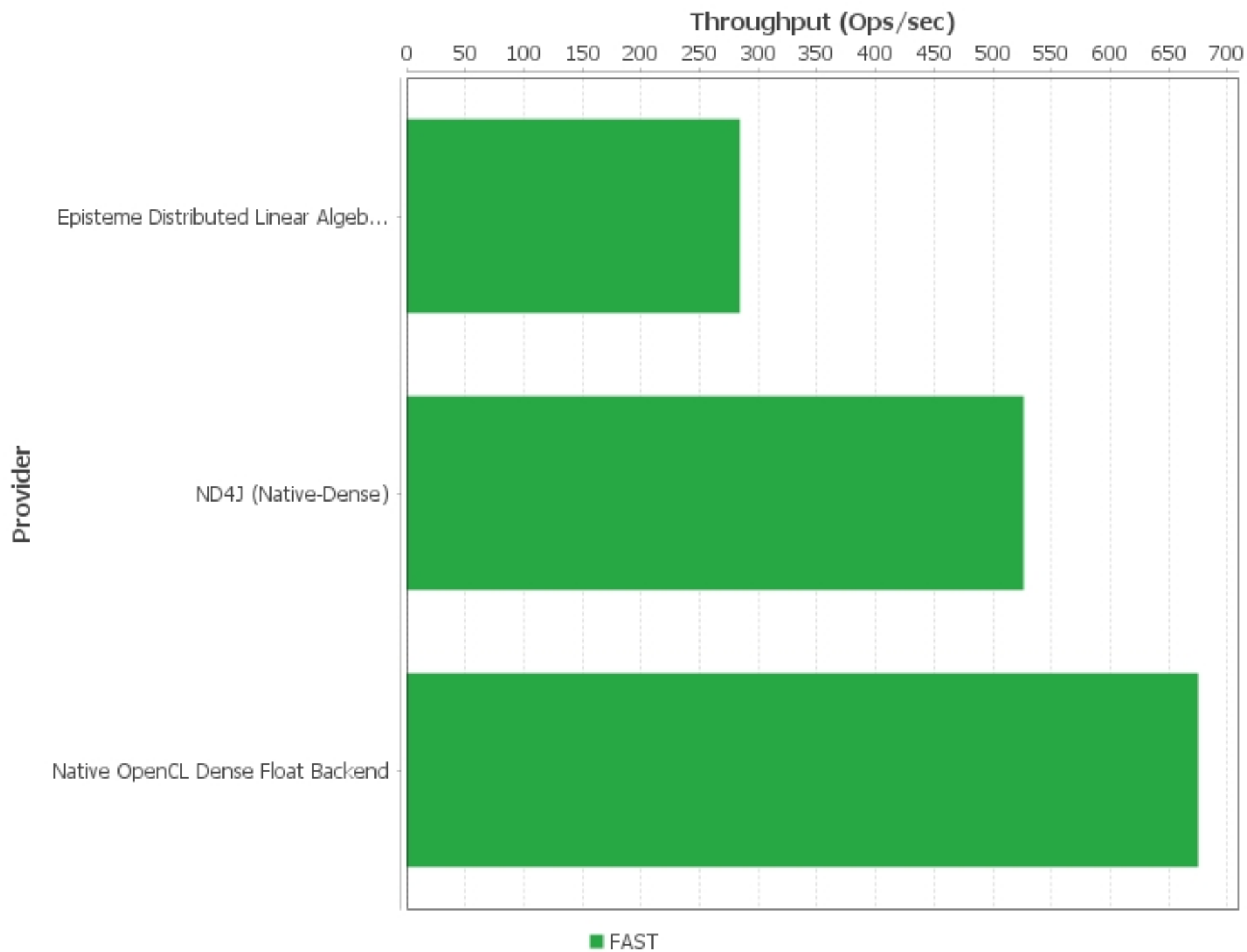
R:MatVec Performance Comparison



Operation Comparison: R:Trans (FAST)

Episteme Multimodal Audit | R:Trans

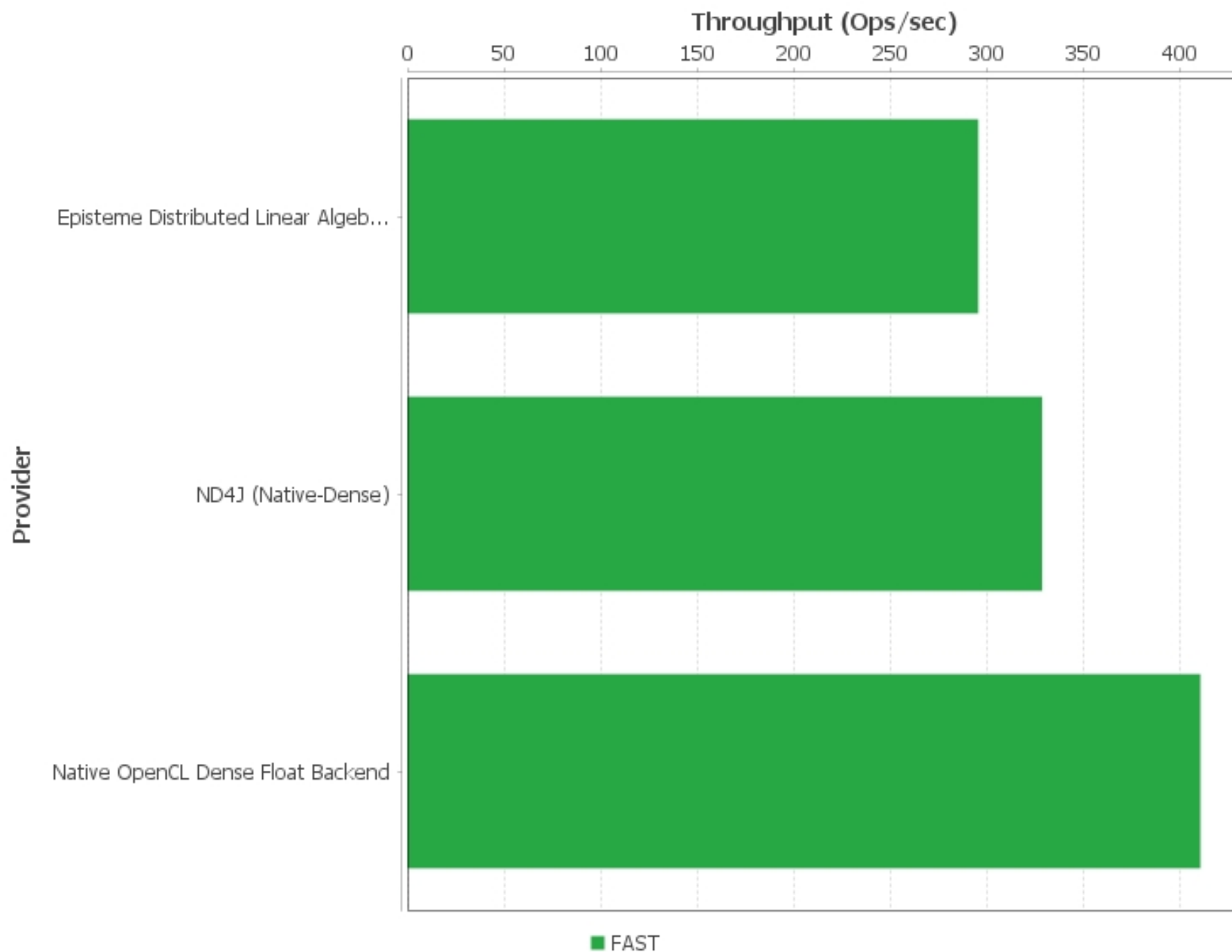
R:Trans Performance Comparison



Operation Comparison: R:Inv (FAST)

Episteme Multimodal Audit | R:Inv

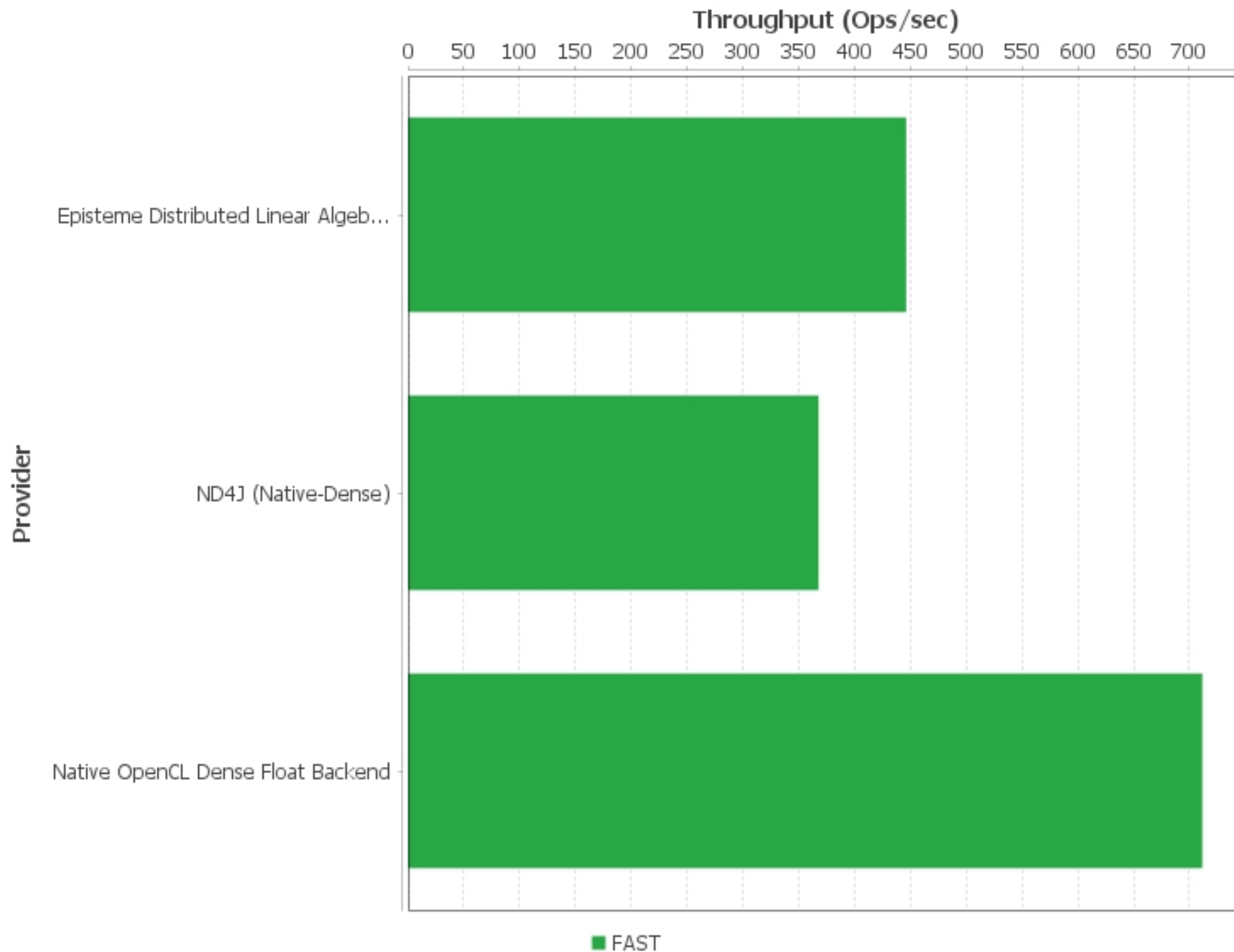
R:Inv Performance Comparison



Operation Comparison: R:Det (FAST)

Episteme Multimodal Audit | R:Det

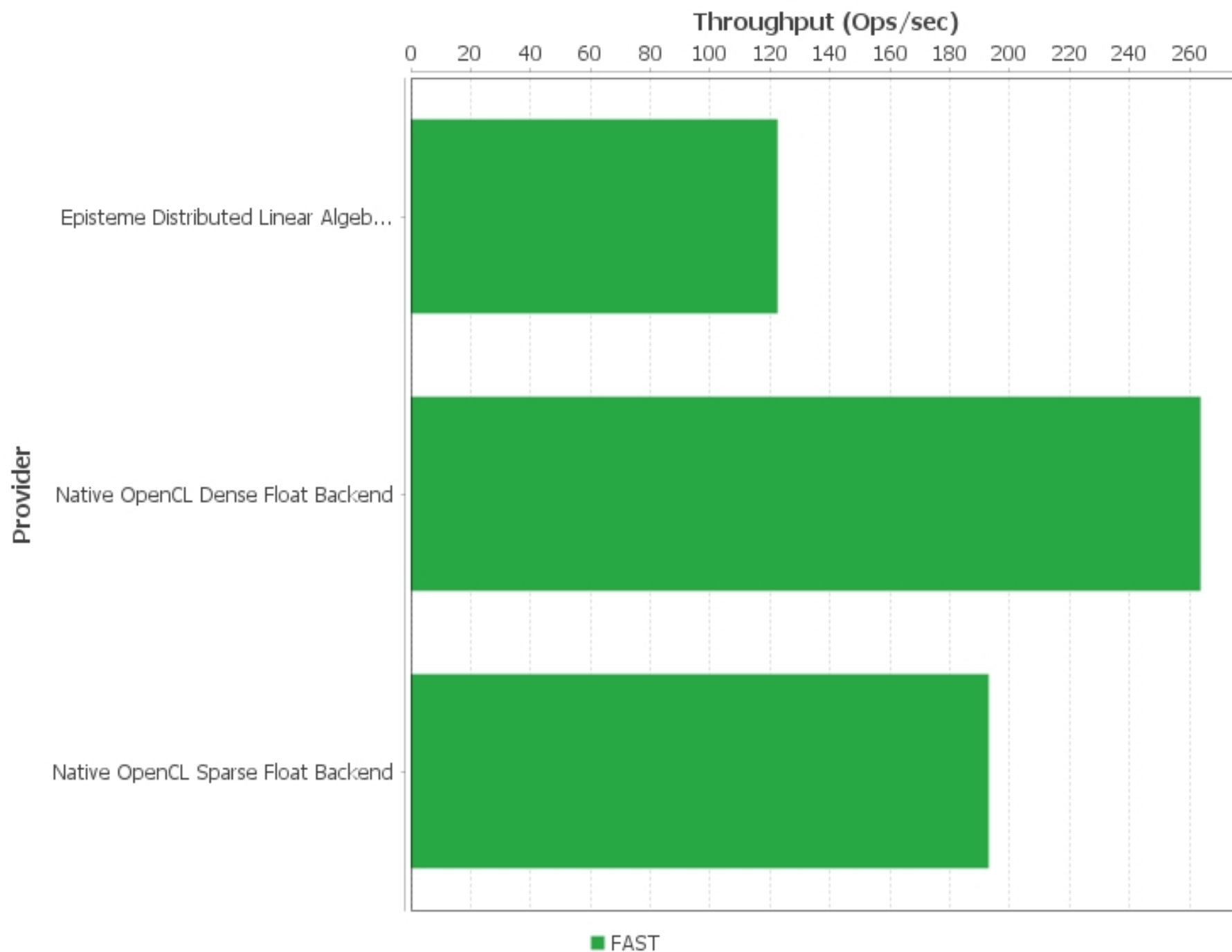
R:Det Performance Comparison



Operation Comparison: R:Solve (FAST)

Episteme Multimodal Audit | R:Solve

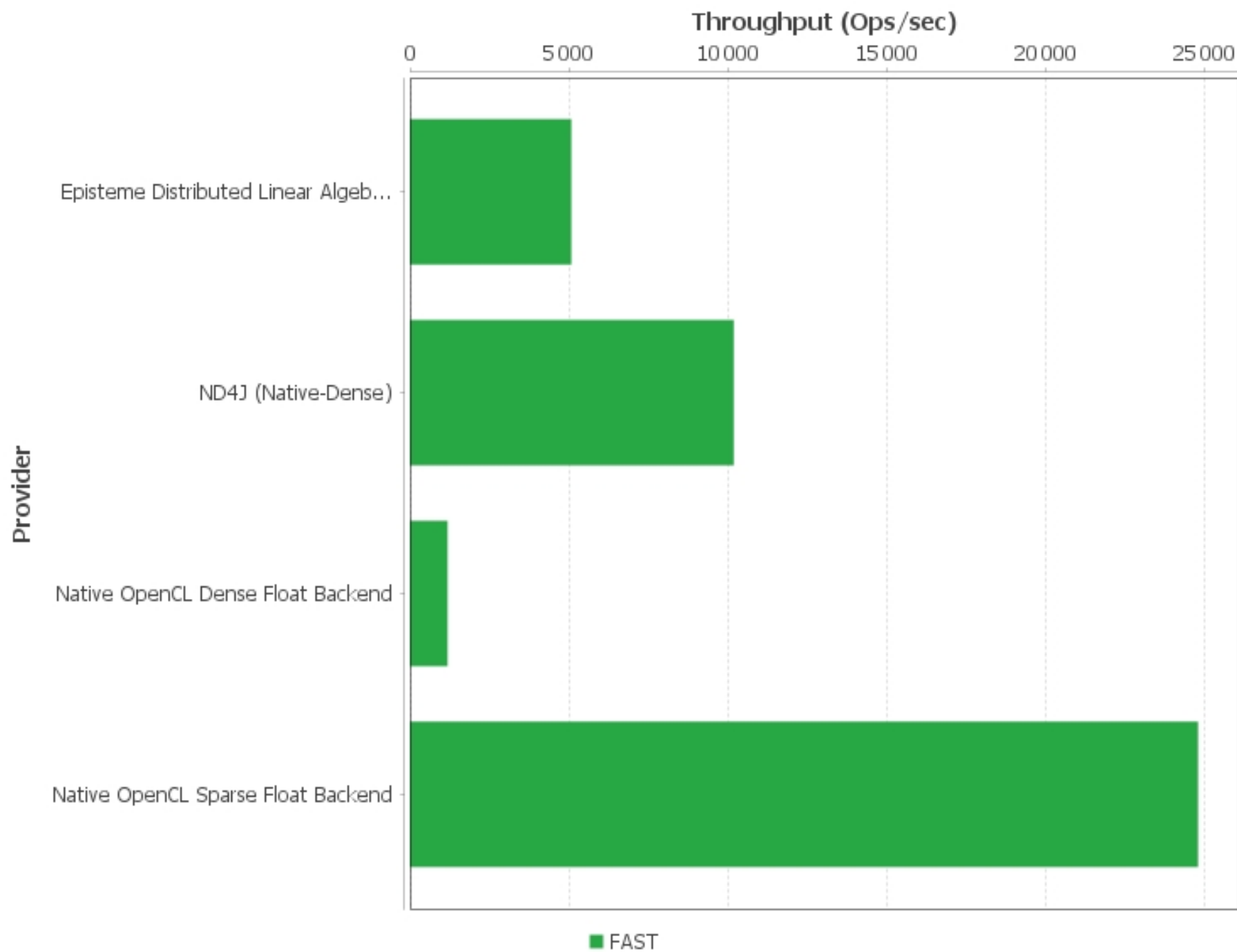
R:Solve Performance Comparison



Operation Comparison: R:Trace (FAST)

Episteme Multimodal Audit | R:Trace

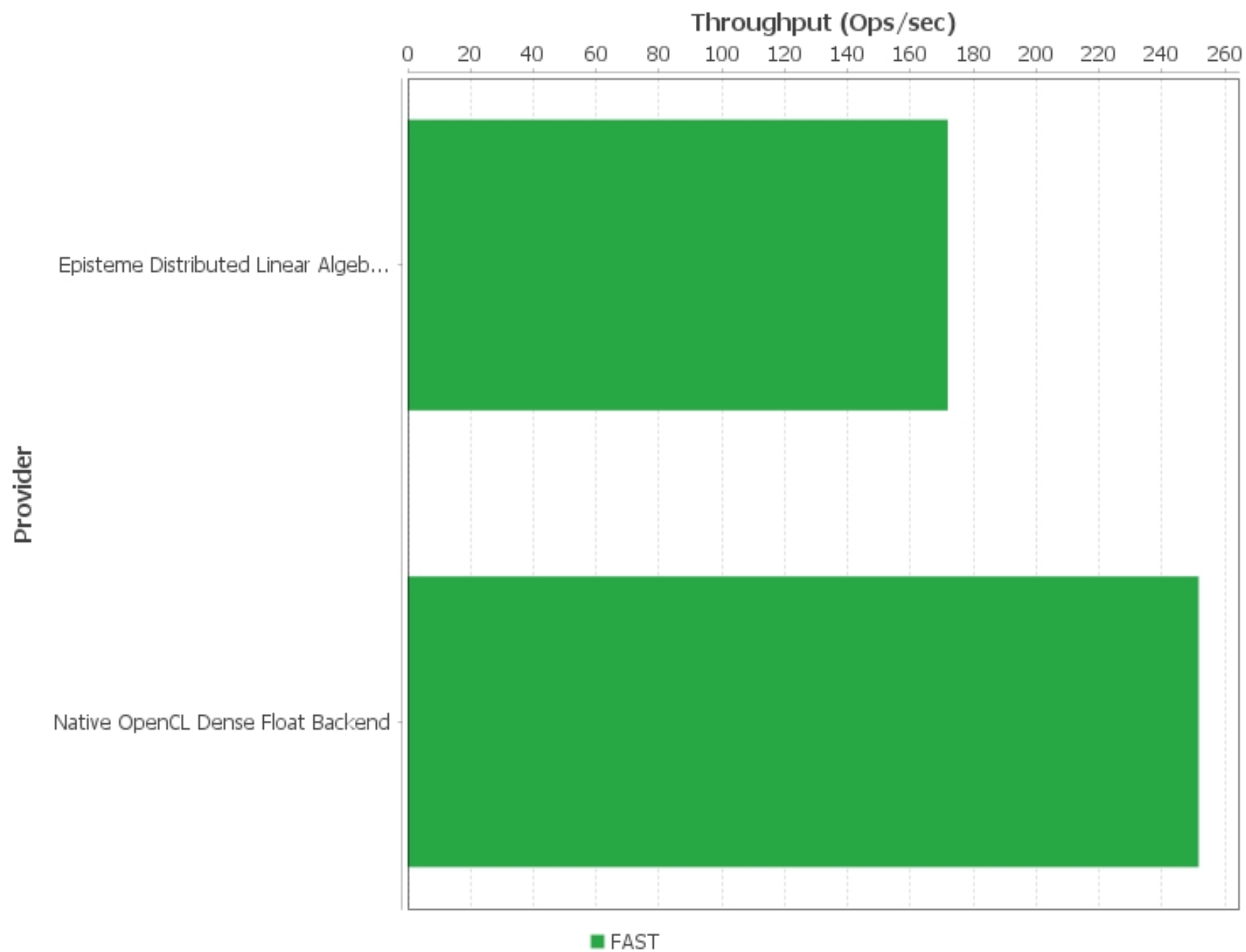
R:Trace Performance Comparison



Operation Comparison: R:LU (FAST)

Episteme Multimodal Audit | R:LU

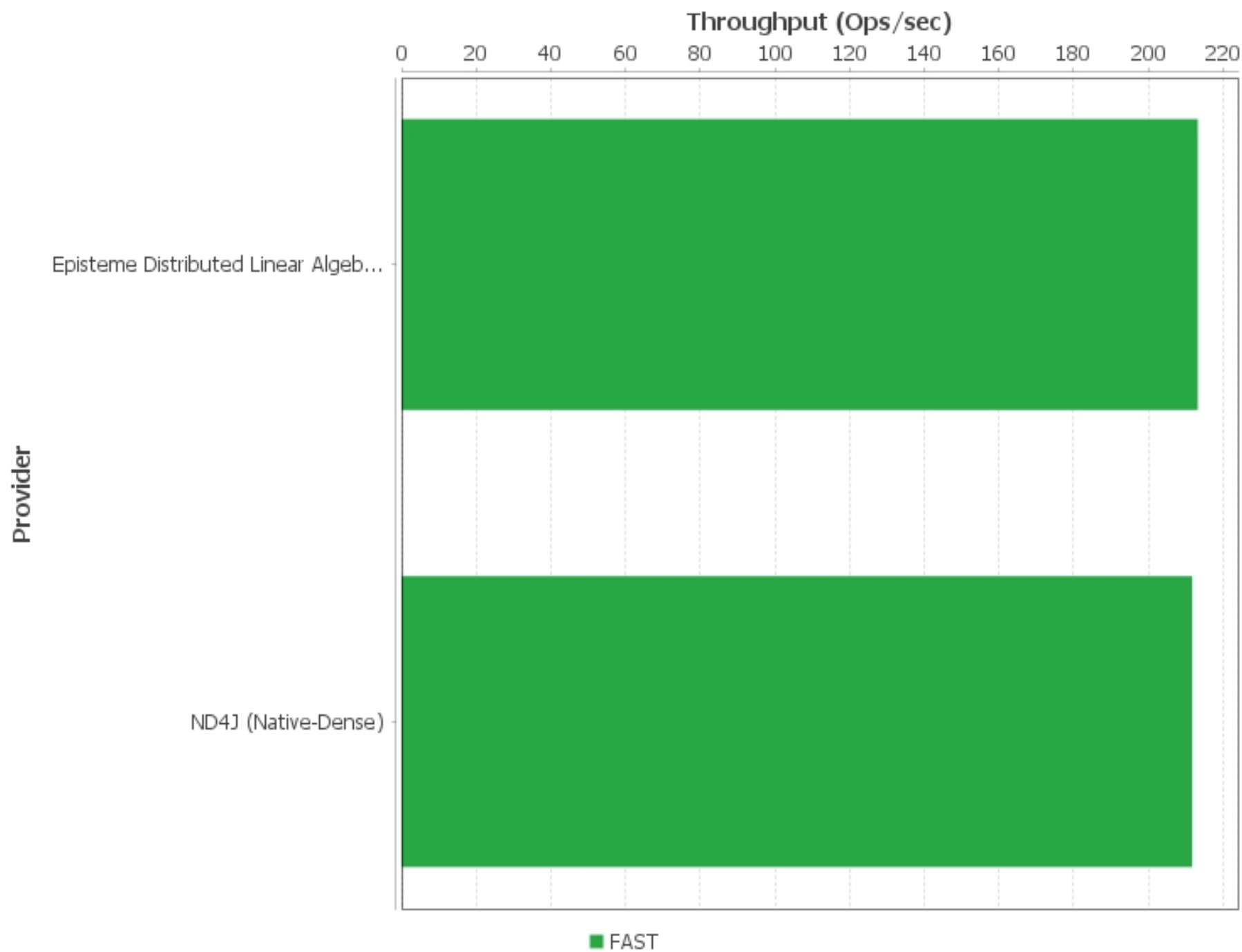
R:LU Performance Comparison



Operation Comparison: R:QR (FAST)

Episteme Multimodal Audit | R:QR

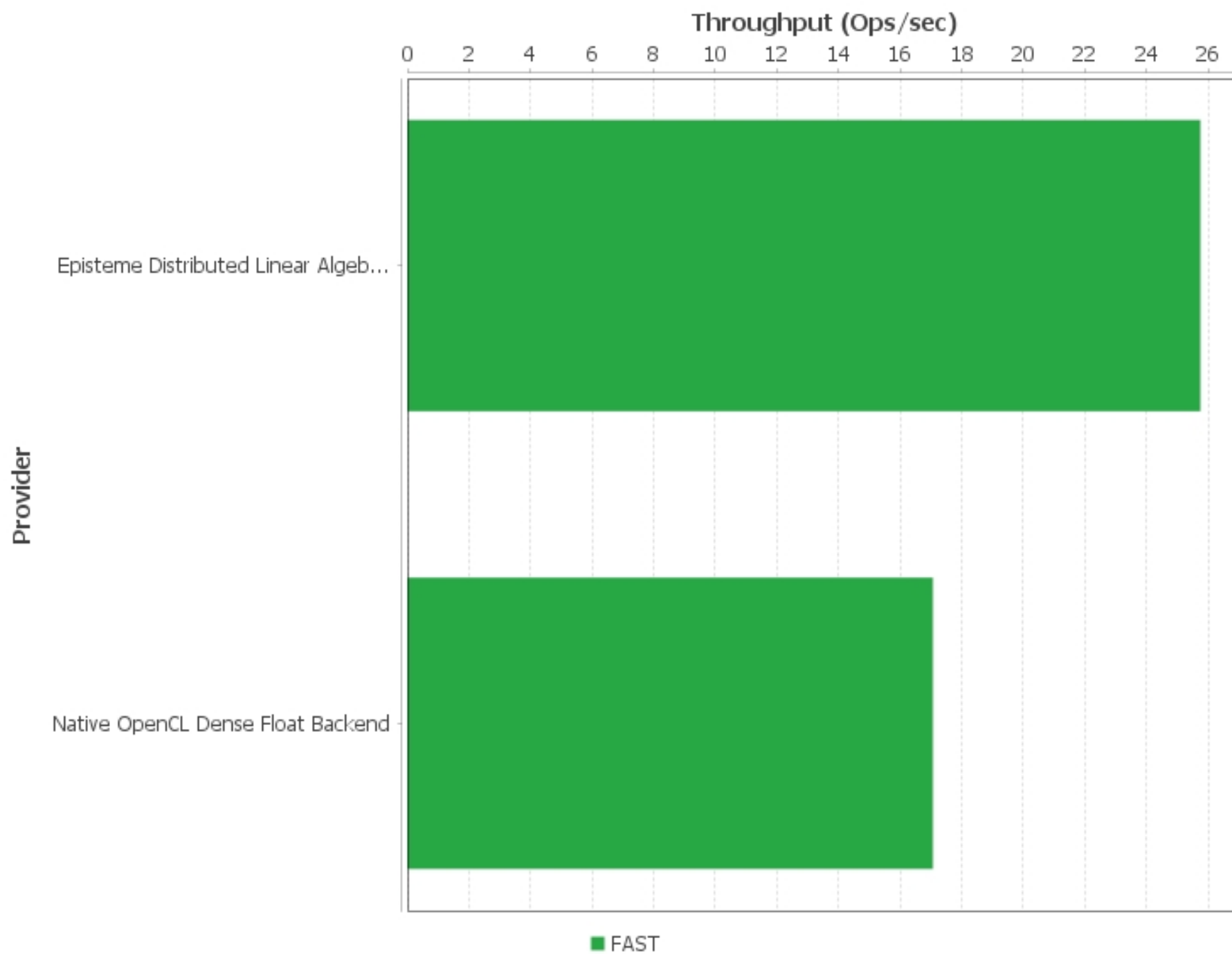
R:QR Performance Comparison



Operation Comparison: R:SVD (FAST)

Episteme Multimodal Audit | R:SVD

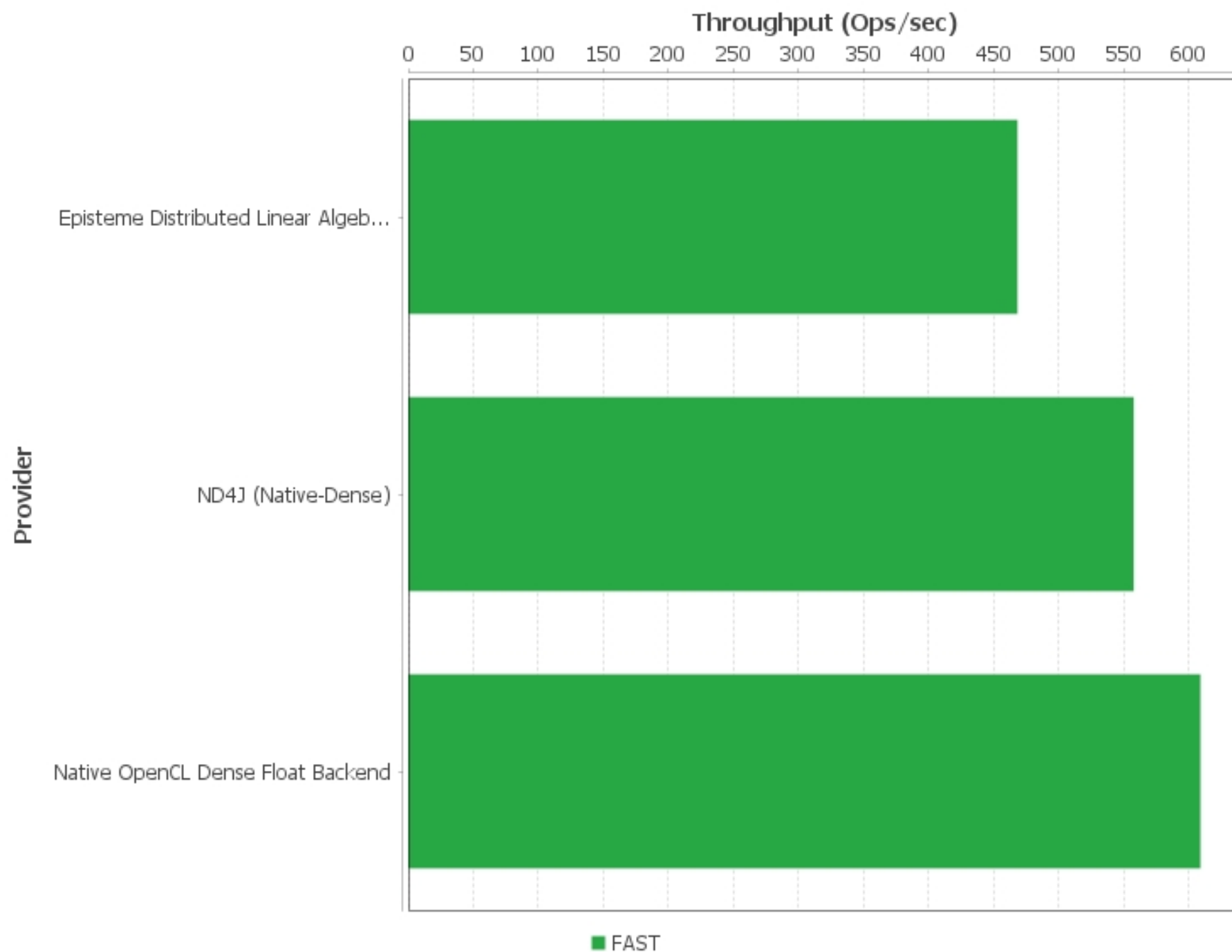
R:SVD Performance Comparison



Operation Comparison: R:Chol (FAST)

Episteme Multimodal Audit | R:Chol

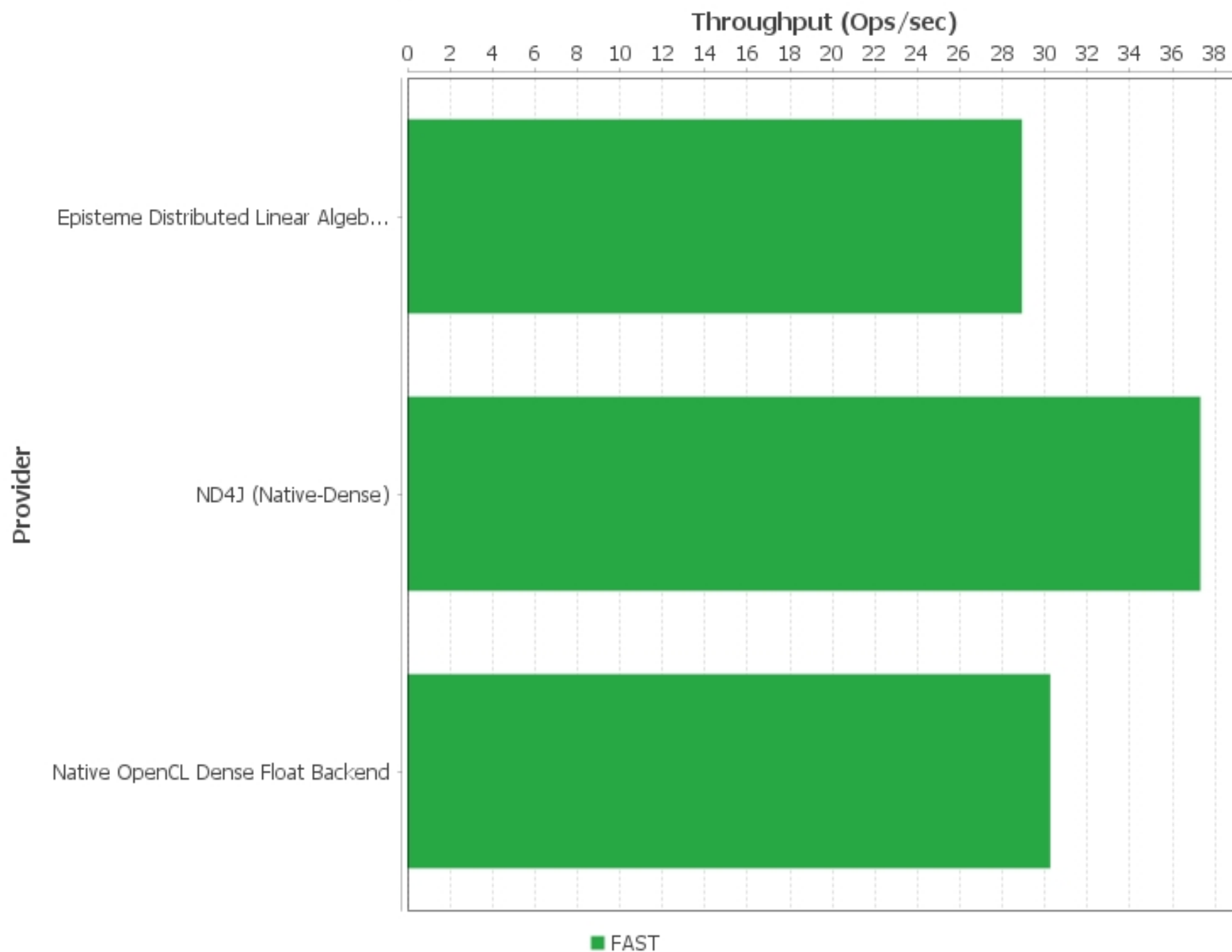
R:Chol Performance Comparison



Operation Comparison: R:Eigen (FAST)

Episteme Multimodal Audit | R:Eigen

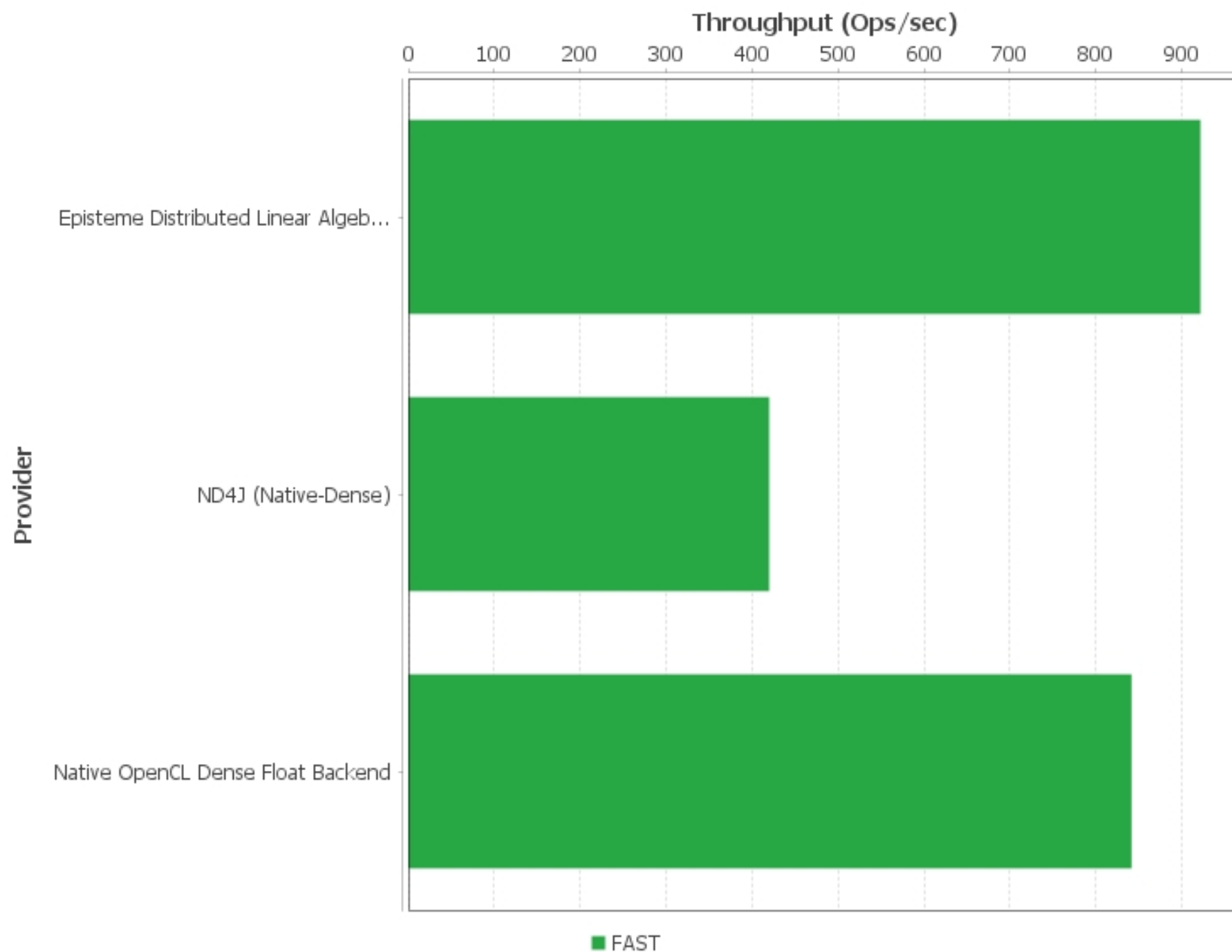
R:Eigen Performance Comparison



Operation Comparison: R:Rect:Mul (FAST)

Episteme Multimodal Audit | R:Rect:Mul

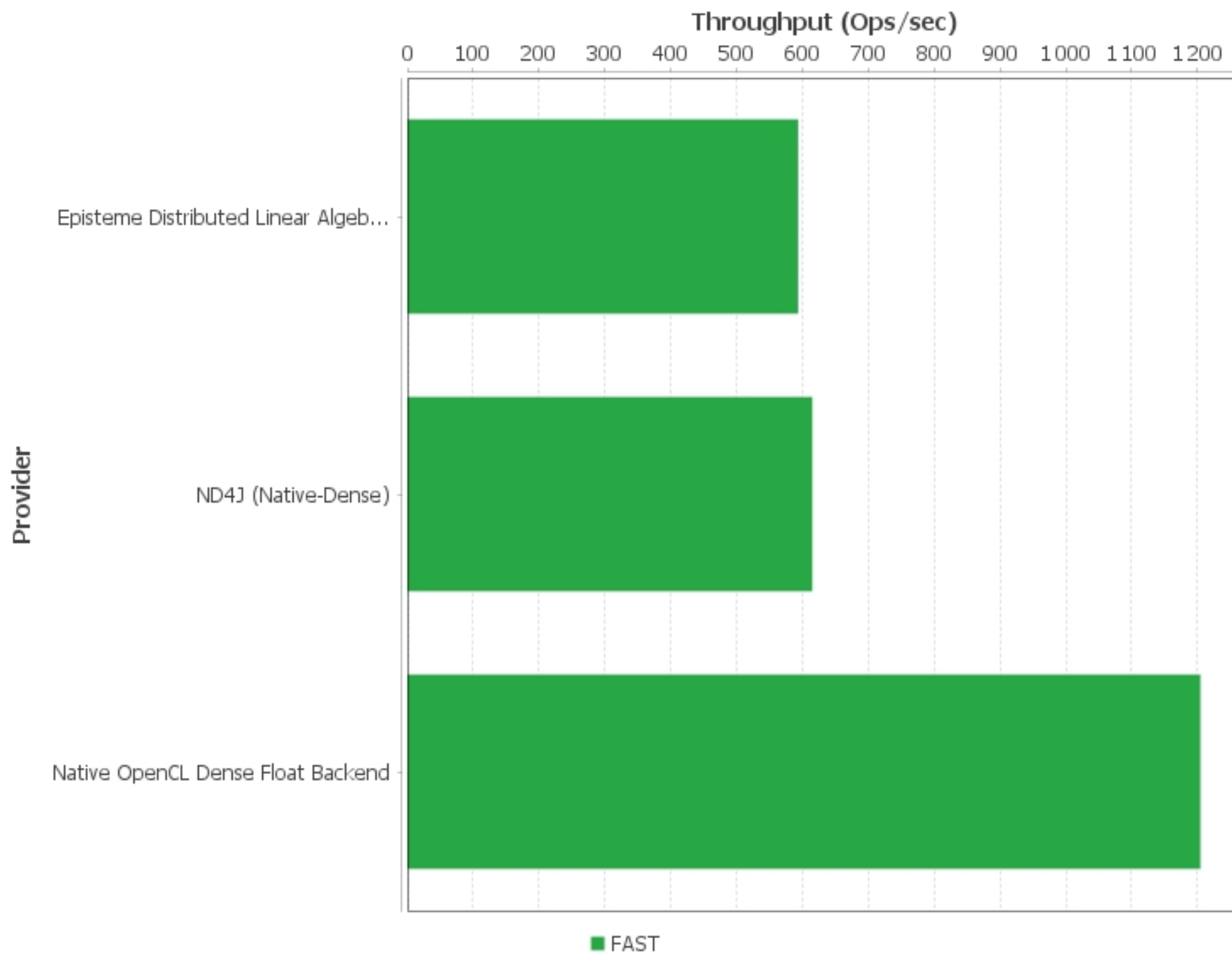
R:Rect:Mul Performance Comparison



Operation Comparison: R:Rect:Trans (FAST)

Episteme Multimodal Audit | R:Rect:Trans

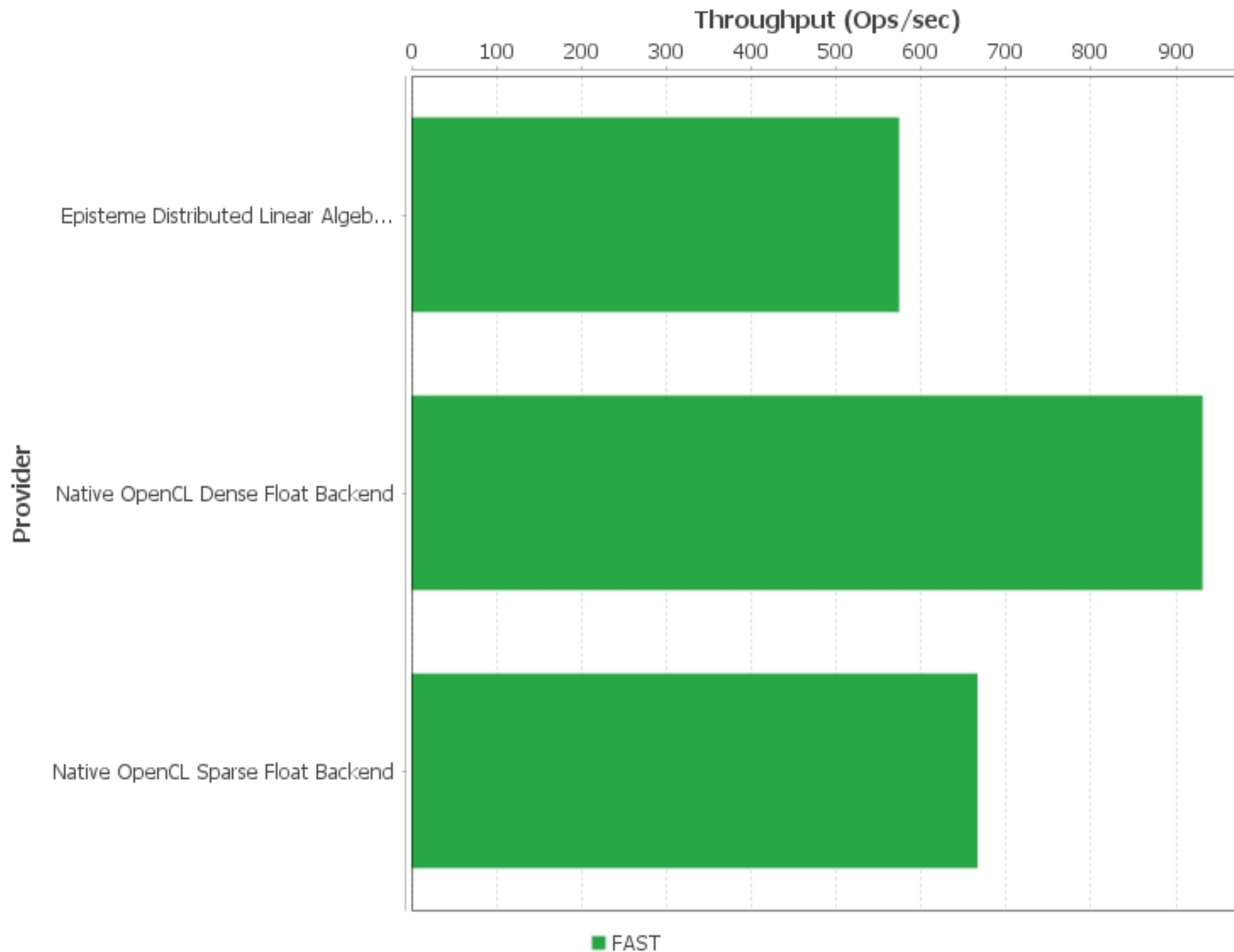
R:Rect:Trans Performance Comparison



Operation Comparison: R:Rect:MatVec (FAST)

Episteme Multimodal Audit | R:Rect:MatVec

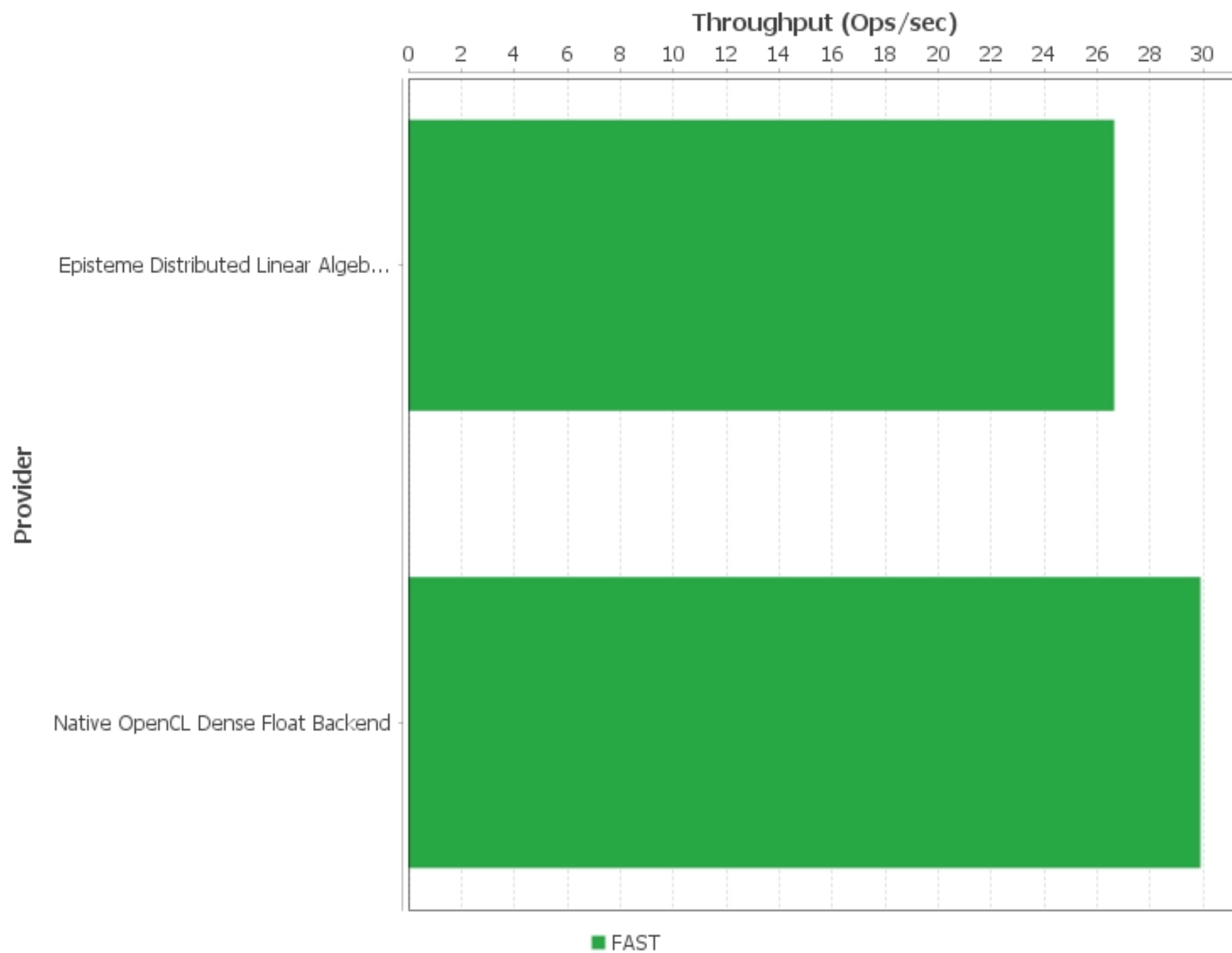
R:Rect:MatVec Performance Comparison



Operation Comparison: R:Rect:SVD (FAST)

Episteme Multimodal Audit | R:Rect:SVD

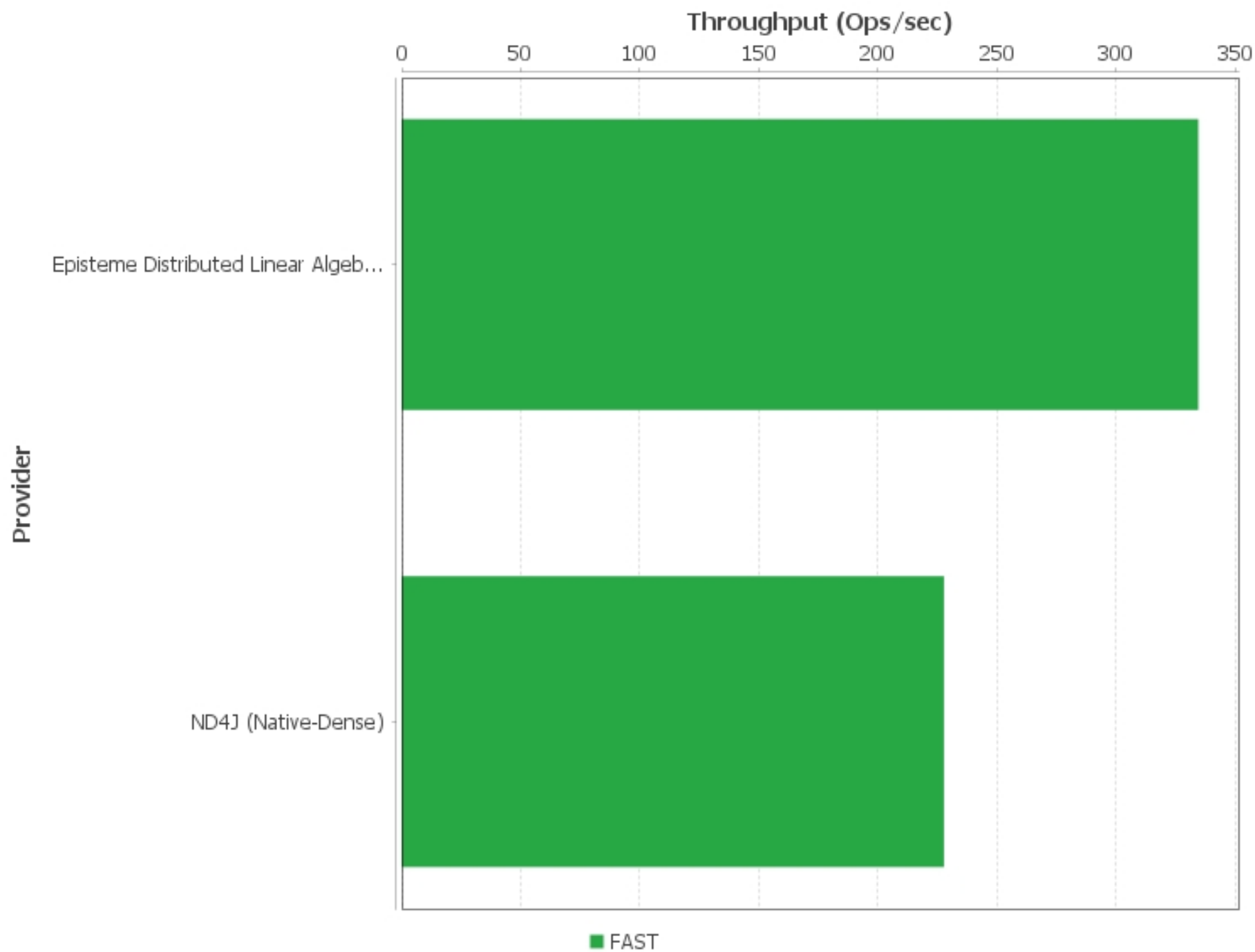
R:Rect:SVD Performance Comparison



Operation Comparison: R:Rect:QR (FAST)

Episteme Multimodal Audit | R:Rect:QR

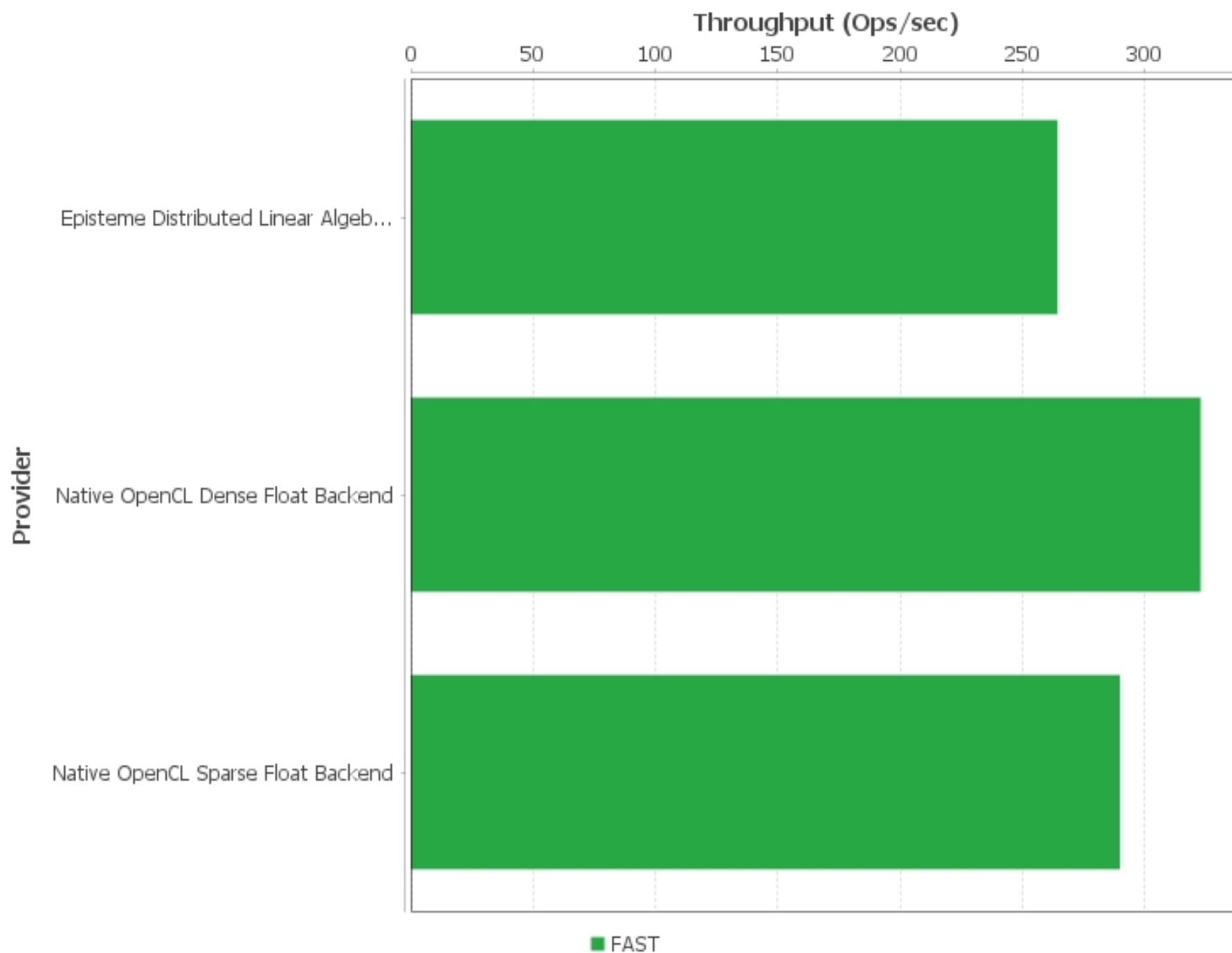
R:Rect:QR Performance Comparison



Operation Comparison: R:Tri:LowerSolve (FAST)

Episteme Multimodal Audit | R:Tri:LowerSolve

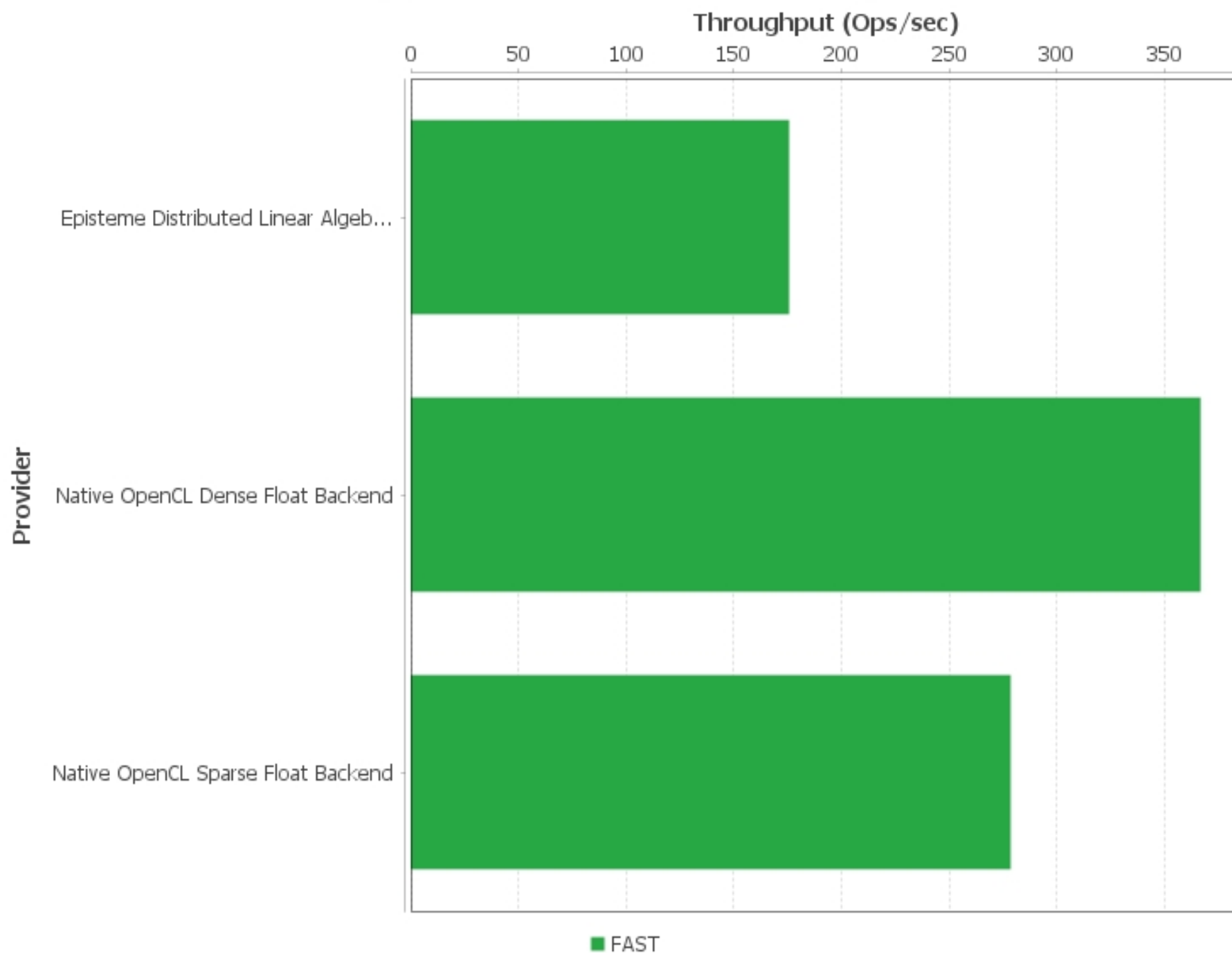
R:Tri:LowerSolve Performance Comparison



Operation Comparison: R:Tri:UpperSolve (FAST)

Episteme Multimodal Audit | R:Tri:UpperSolve

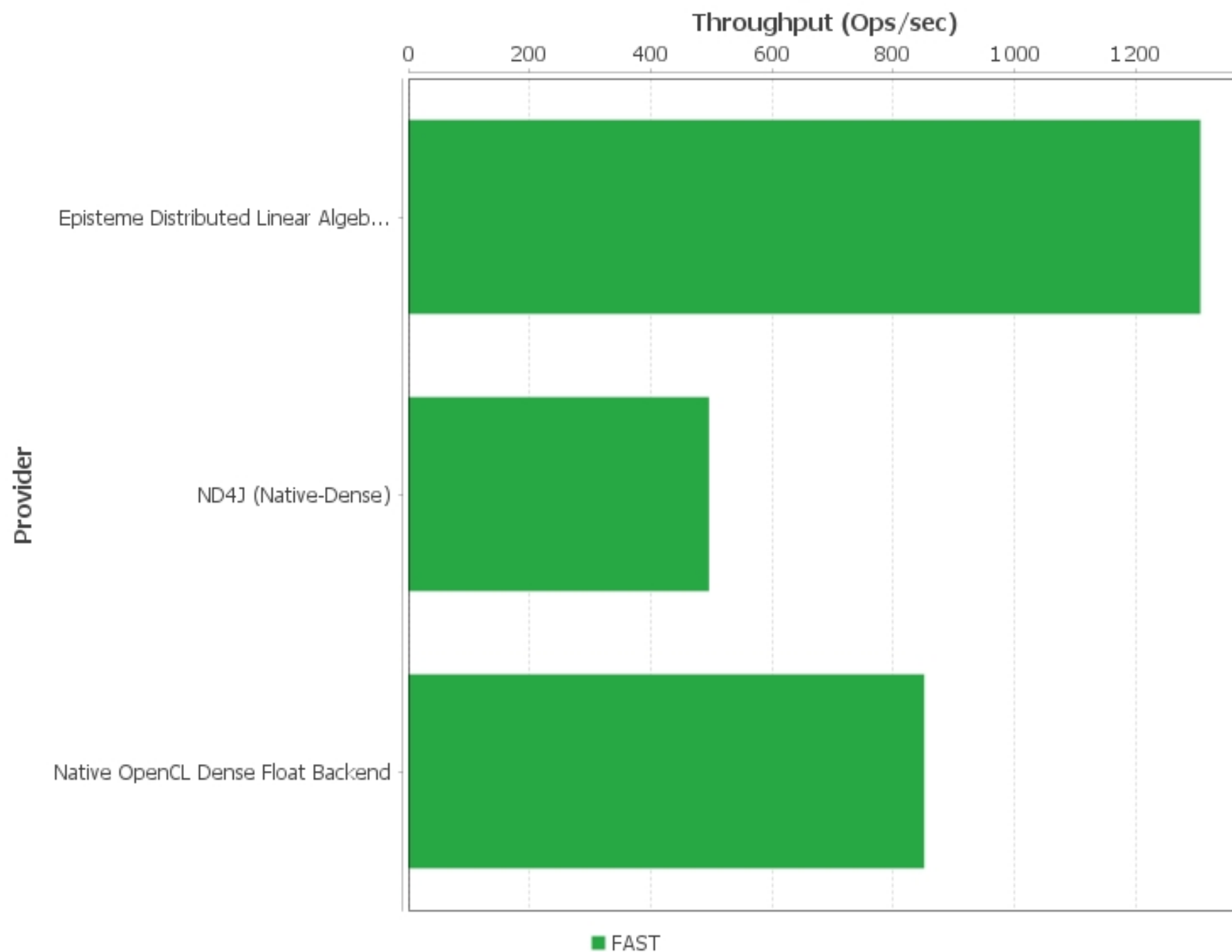
R:Tri:UpperSolve Performance Comparison



Operation Comparison: R:Tri:Mul (FAST)

Episteme Multimodal Audit | R:Tri:Mul

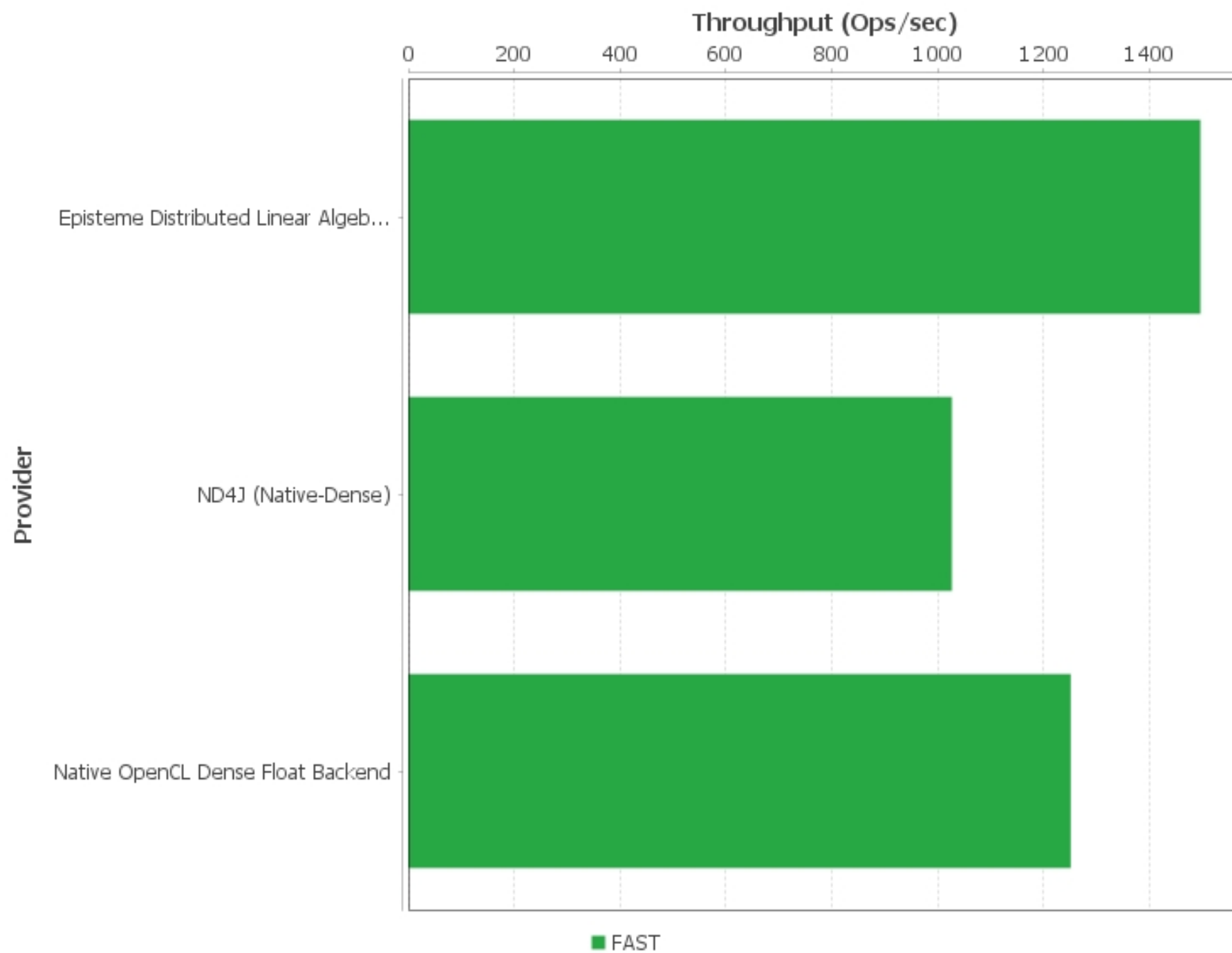
R:Tri:Mul Performance Comparison



Operation Comparison: R:Vec:Dot (FAST)

Episteme Multimodal Audit | R:Vec:Dot

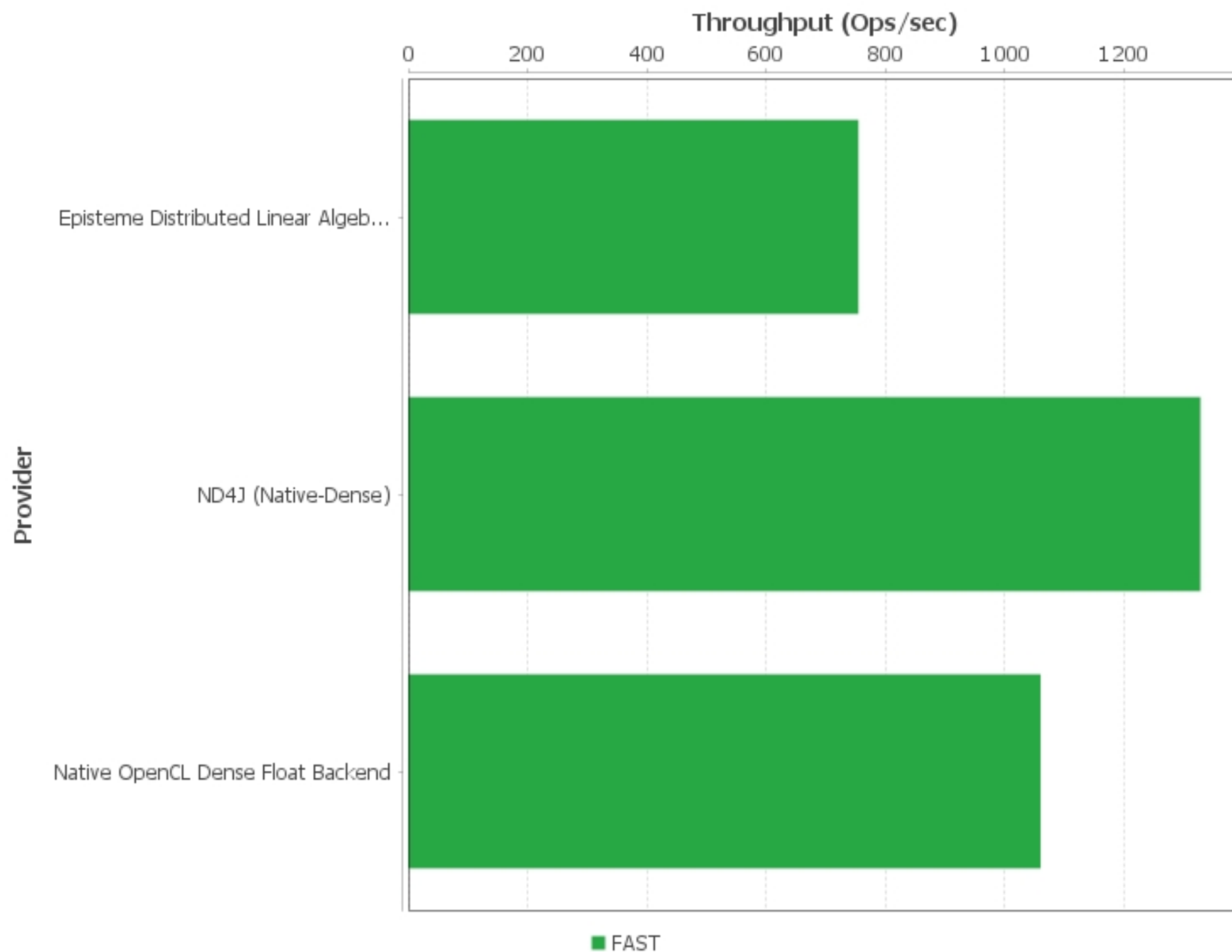
R:Vec:Dot Performance Comparison



Operation Comparison: R:Vec:Norm (FAST)

Episteme Multimodal Audit | R:Vec:Norm

R:Vec:Norm Performance Comparison



Operation Comparison: R:Vec:Normalize (FAST)

Operation Comparison: R:Vec:Cross (FAST)

Episteme Multimodal Audit | R:Vec:Cross

Operation Comparison: R:Vec:Angle (FAST)

Episteme Multimodal Audit | R:Vec:Angle

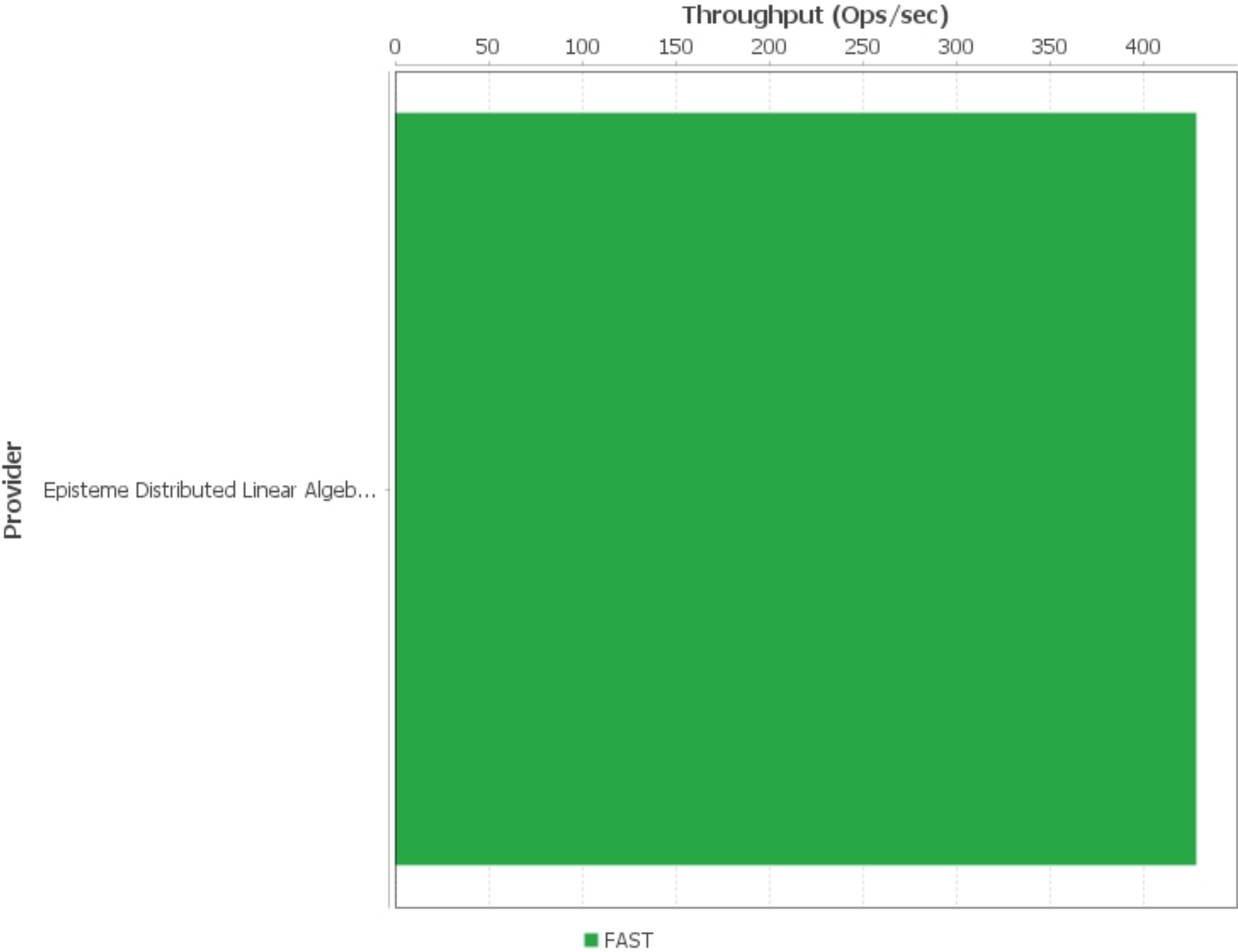
Operation Comparison: R:Vec:Proj (FAST)

Episteme Multimodal Audit | R:Vec:Proj

Operation Comparison: R:Func:Exp (FAST)

Episteme Multimodal Audit | R:Func:Exp

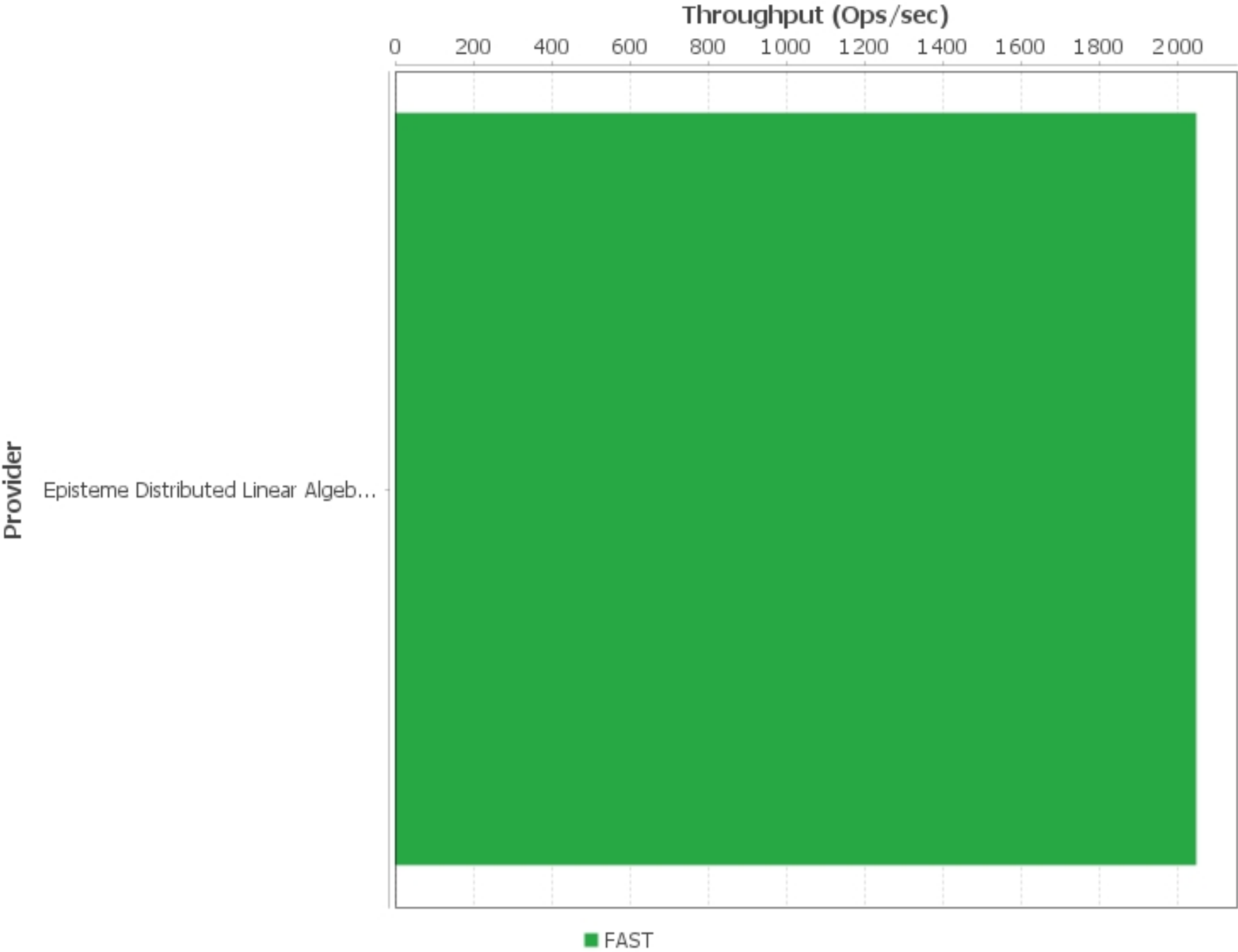
R:Func:Exp Performance Comparison



Operation Comparison: R:Func:Log (FAST)

Episteme Multimodal Audit | R:Func:Log

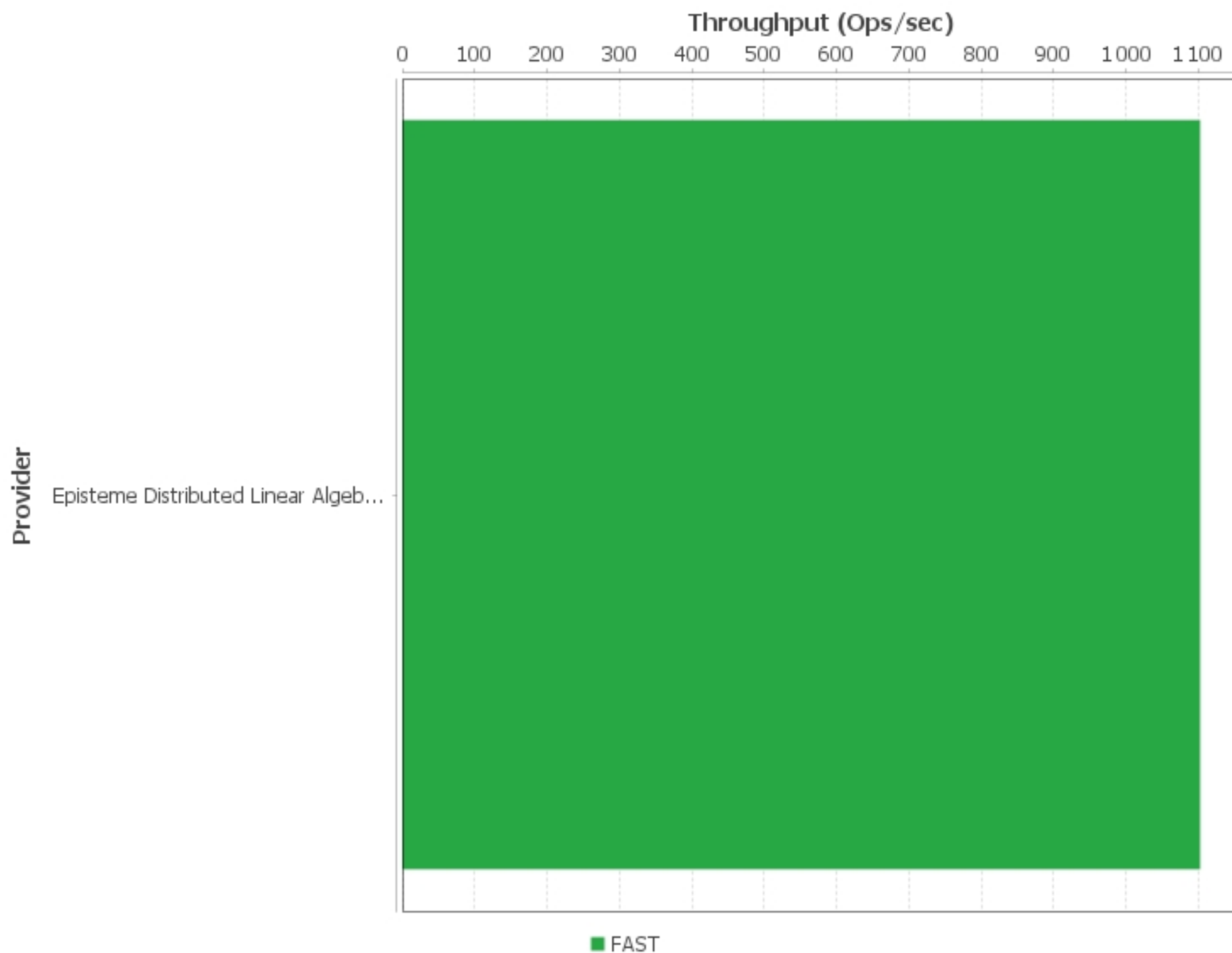
R:Func:Log Performance Comparison



Operation Comparison: R:Func:Sin (FAST)

Episteme Multimodal Audit | R:Func:Sin

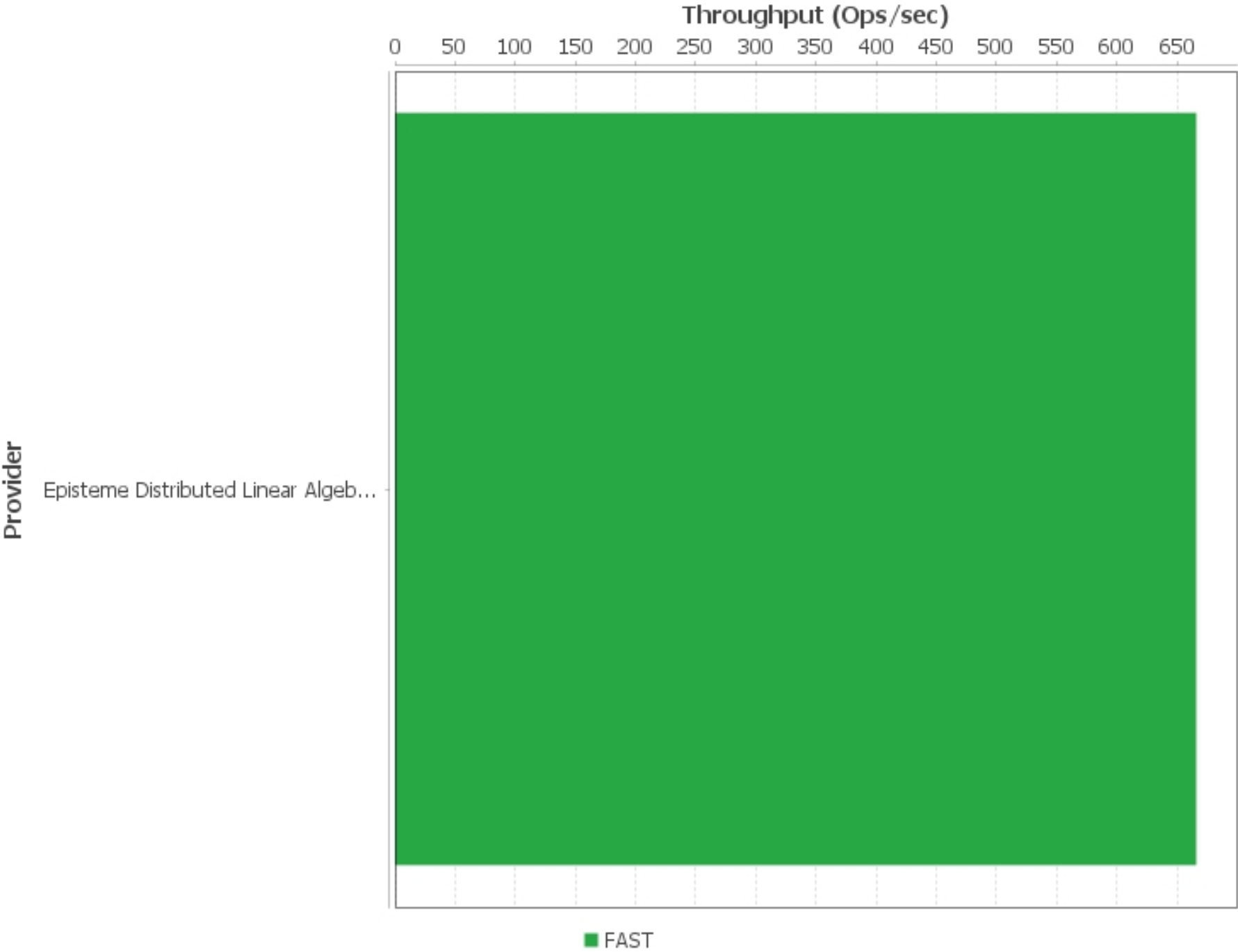
R:Func:Sin Performance Comparison



Operation Comparison: R:Func:Cos (FAST)

Episteme Multimodal Audit | R:Func:Cos

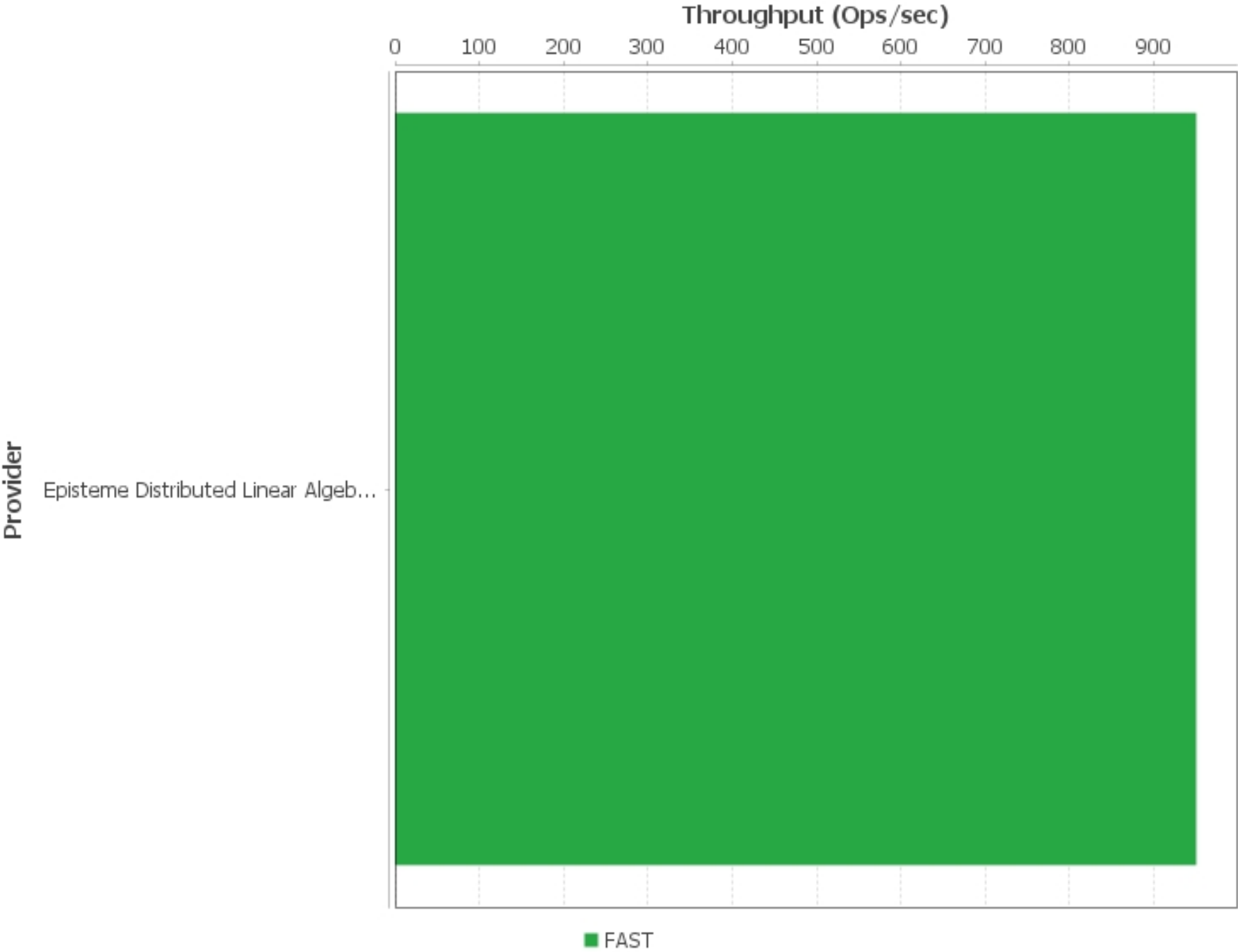
R:Func:Cos Performance Comparison



Operation Comparison: R:Func:Tan (FAST)

Episteme Multimodal Audit | R:Func:Tan

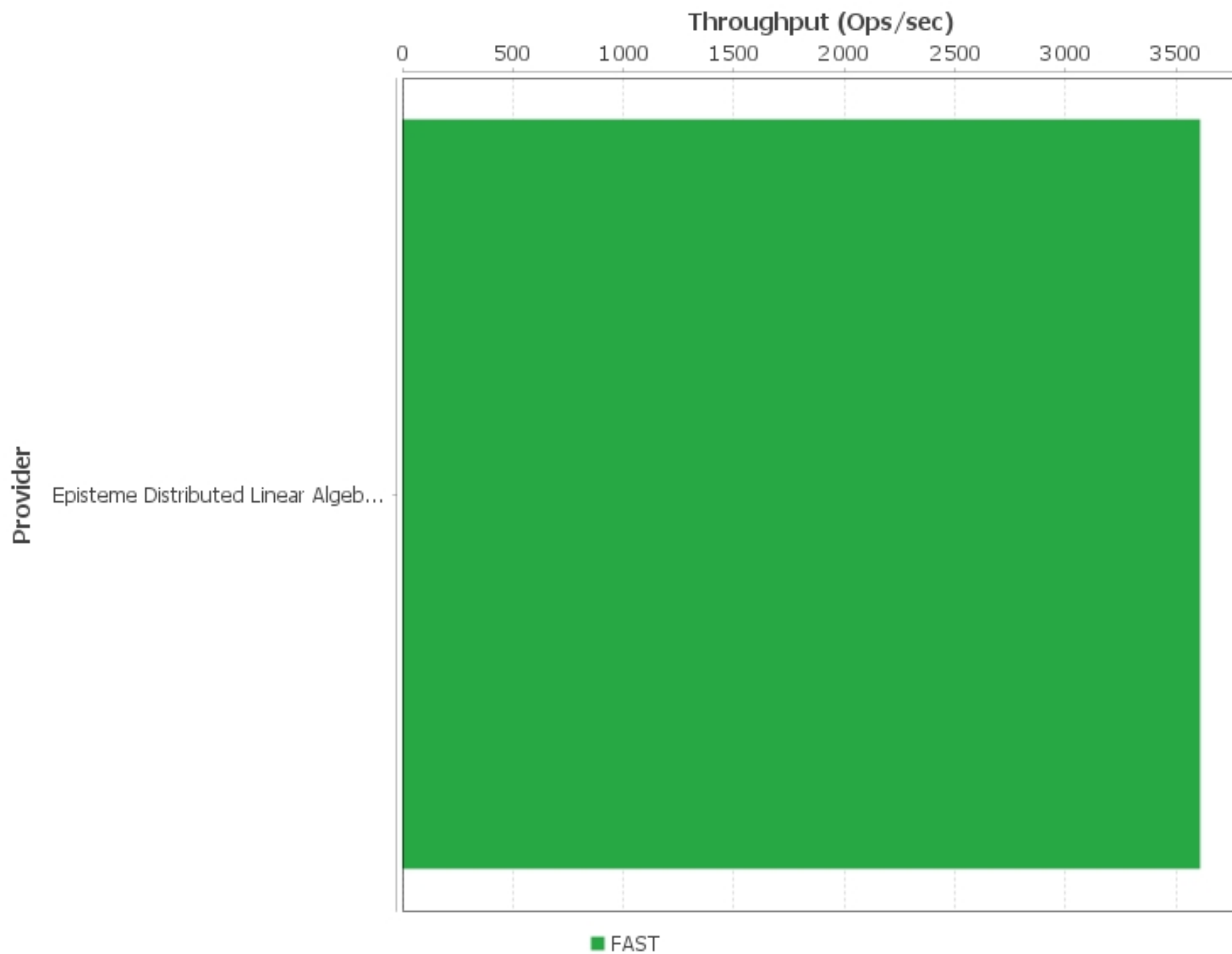
R:Func:Tan Performance Comparison



Operation Comparison: R:Func:Sinh (FAST)

Episteme Multimodal Audit | R:Func:Sinh

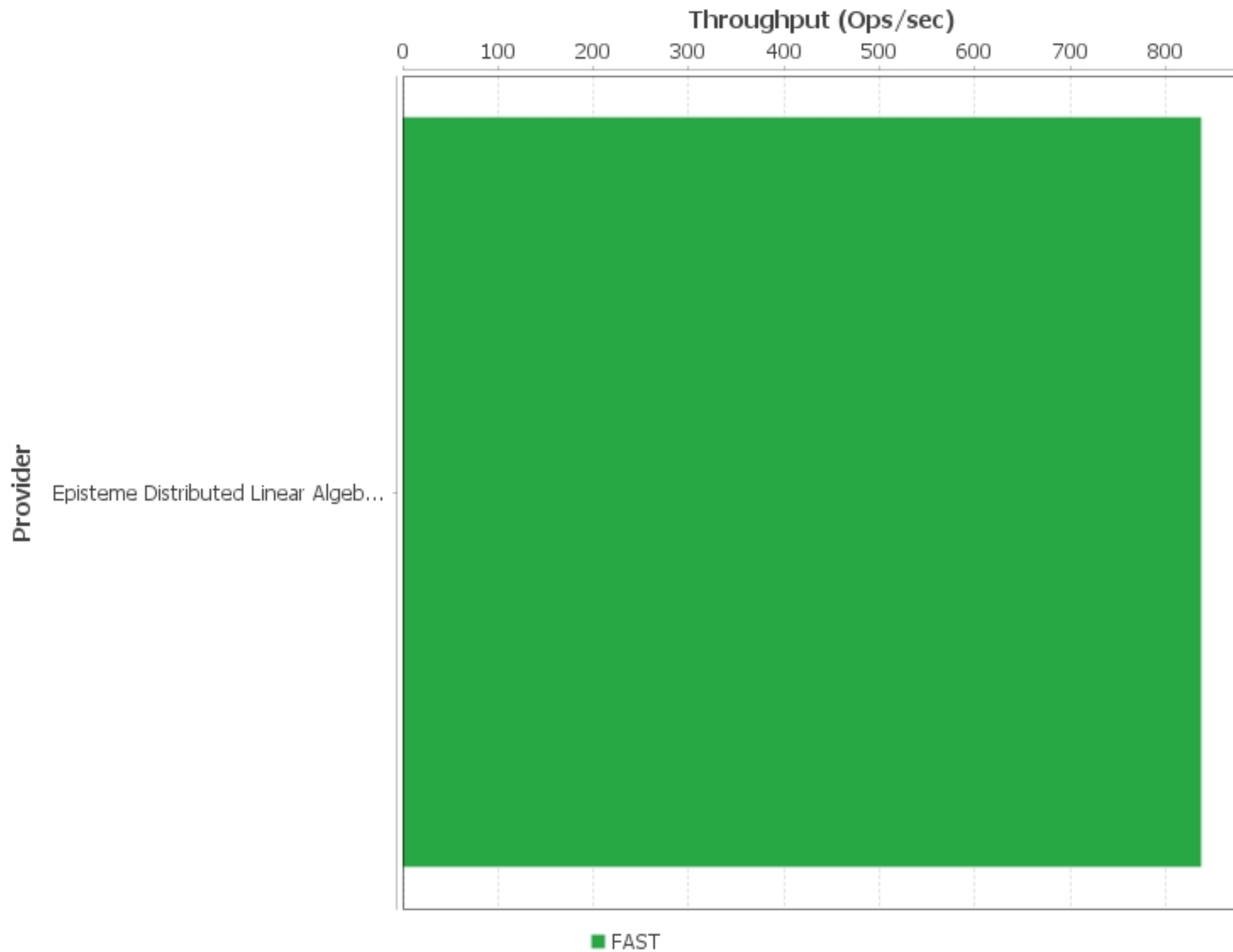
R:Func:Sinh Performance Comparison



Operation Comparison: R:Func:Cosh (FAST)

Episteme Multimodal Audit | R:Func:Cosh

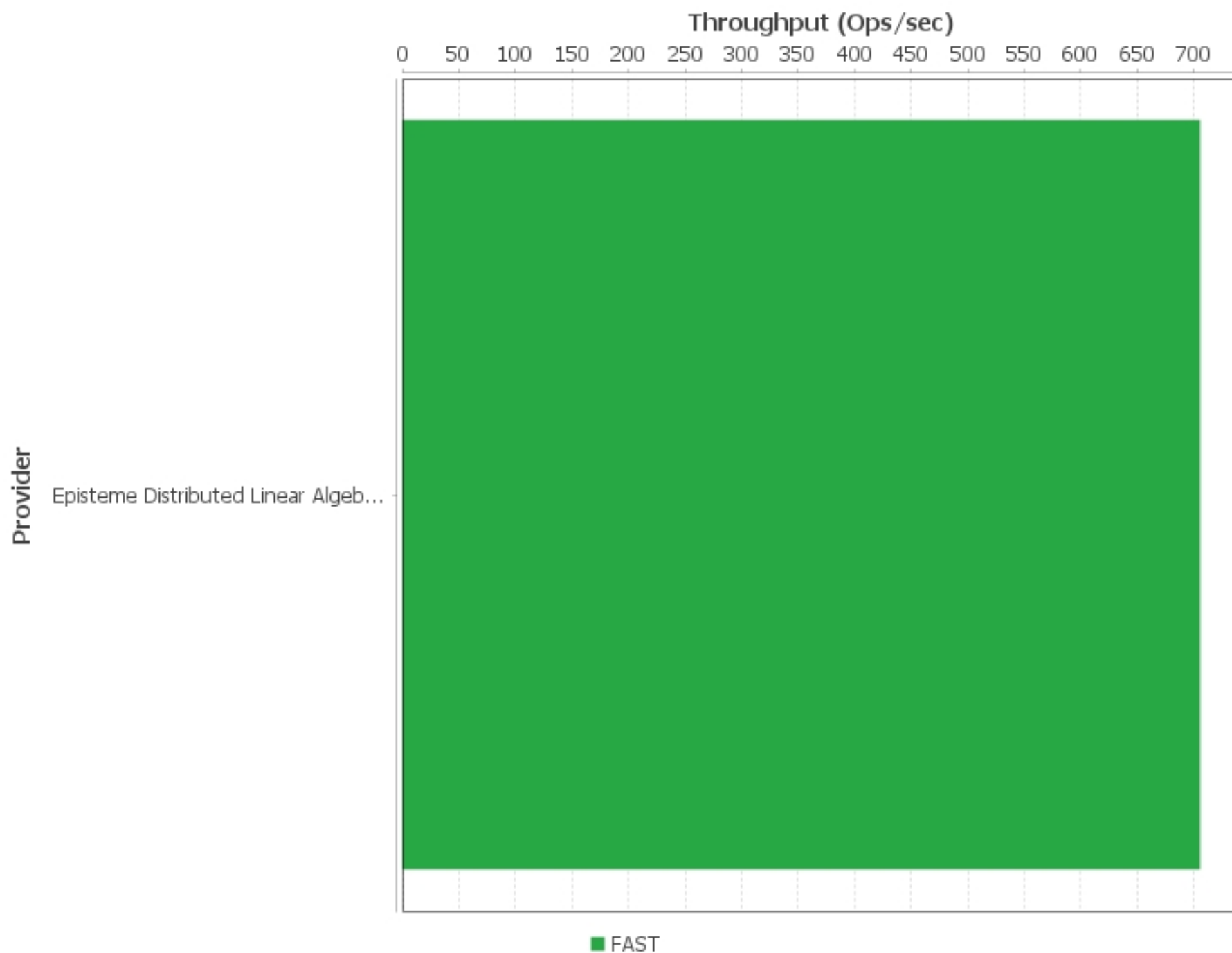
R:Func:Cosh Performance Comparison



Operation Comparison: R:Func:Tanh (FAST)

Episteme Multimodal Audit | R:Func:Tanh

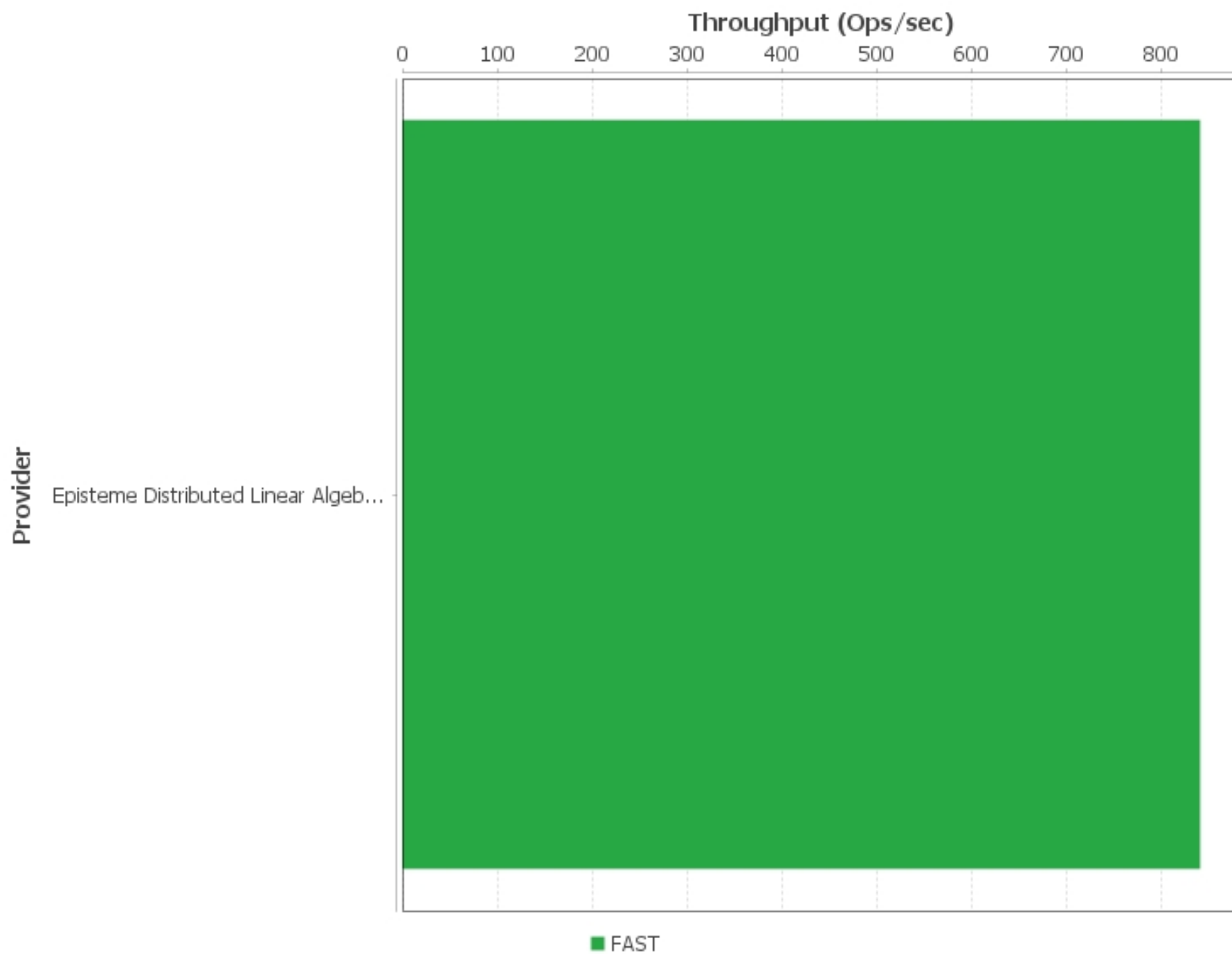
R:Func:Tanh Performance Comparison



Operation Comparison: R:Func:Sqrt (FAST)

Episteme Multimodal Audit | R:Func:Sqrt

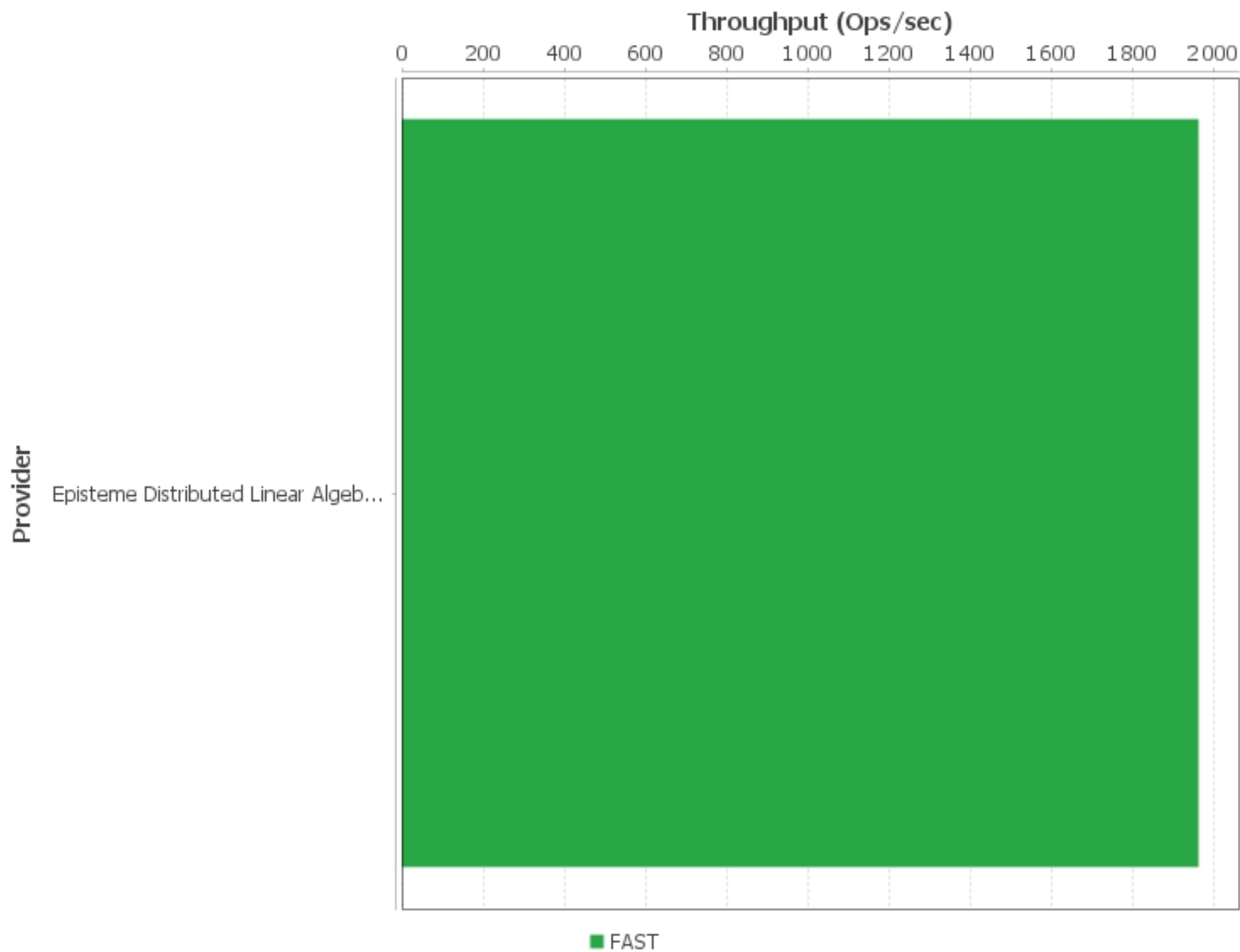
R:Func:Sqrt Performance Comparison



Operation Comparison: R:Func:Cbrt (FAST)

Episteme Multimodal Audit | R:Func:Cbrt

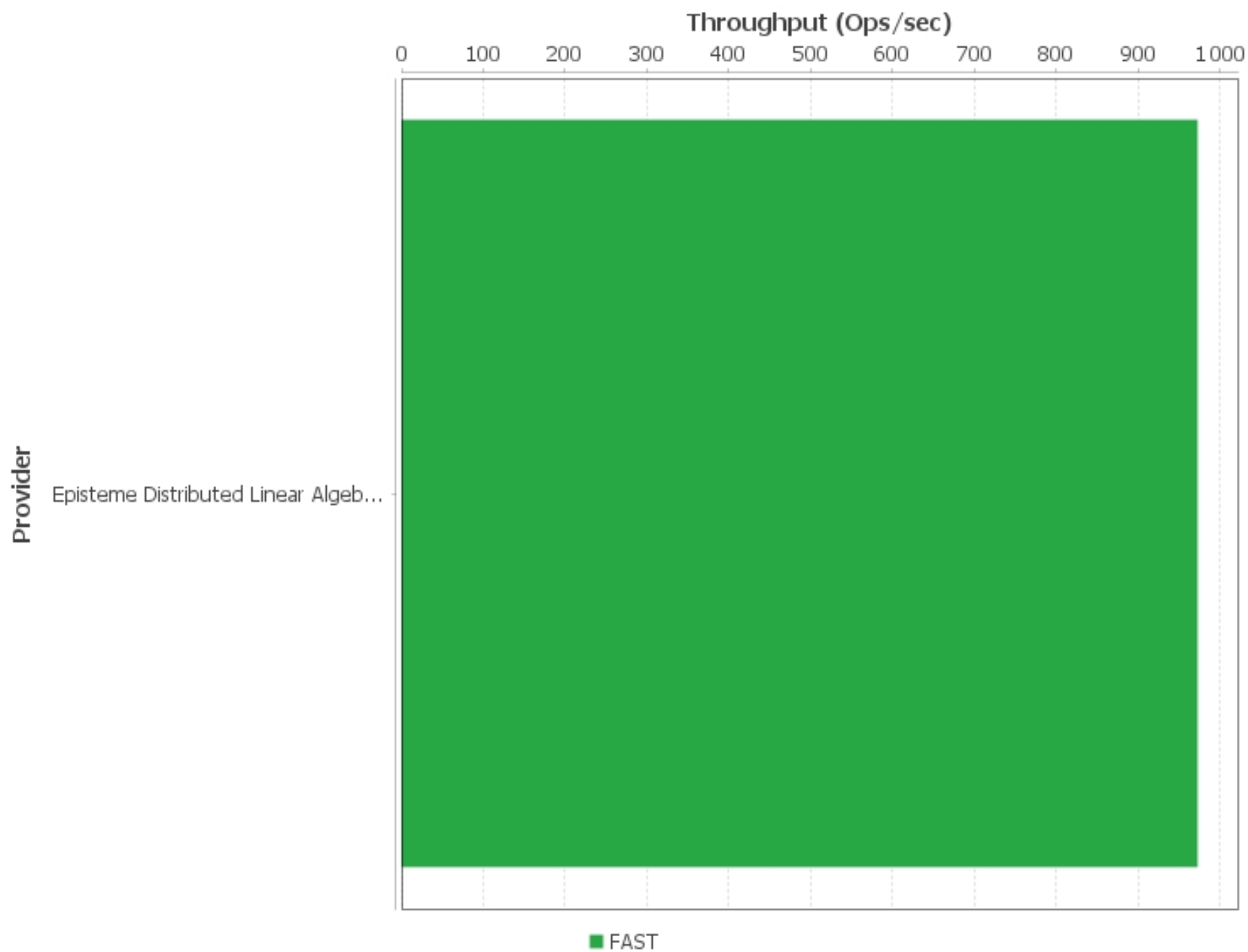
R:Func:Cbrt Performance Comparison



Operation Comparison: R:Func:Pow (FAST)

Episteme Multimodal Audit | R:Func:Pow

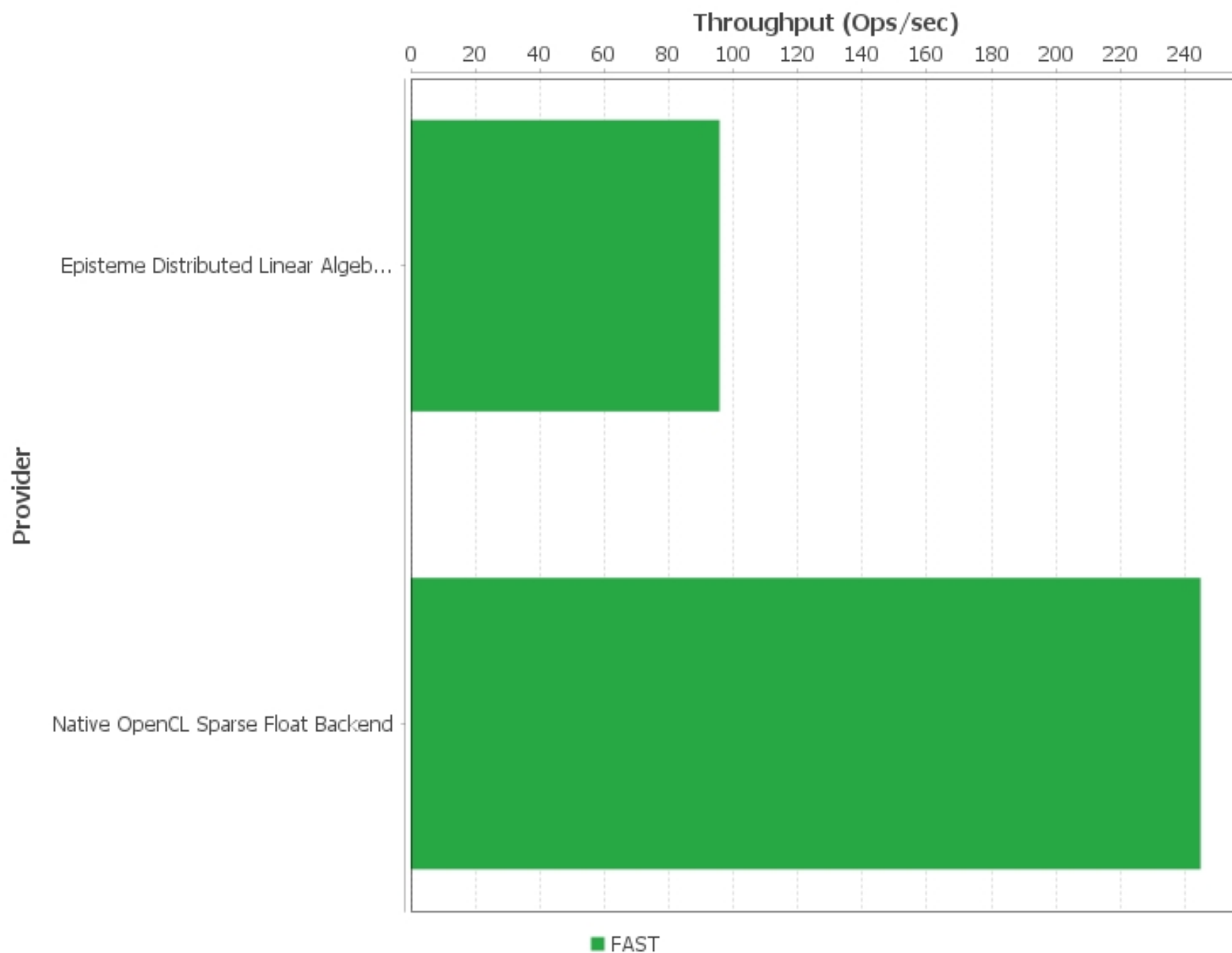
R:Func:Pow Performance Comparison



Operation Comparison: R:Sparse:BiCGSTAB (FAST)

Episteme Multimodal Audit | R:Sparse:BiCGSTAB

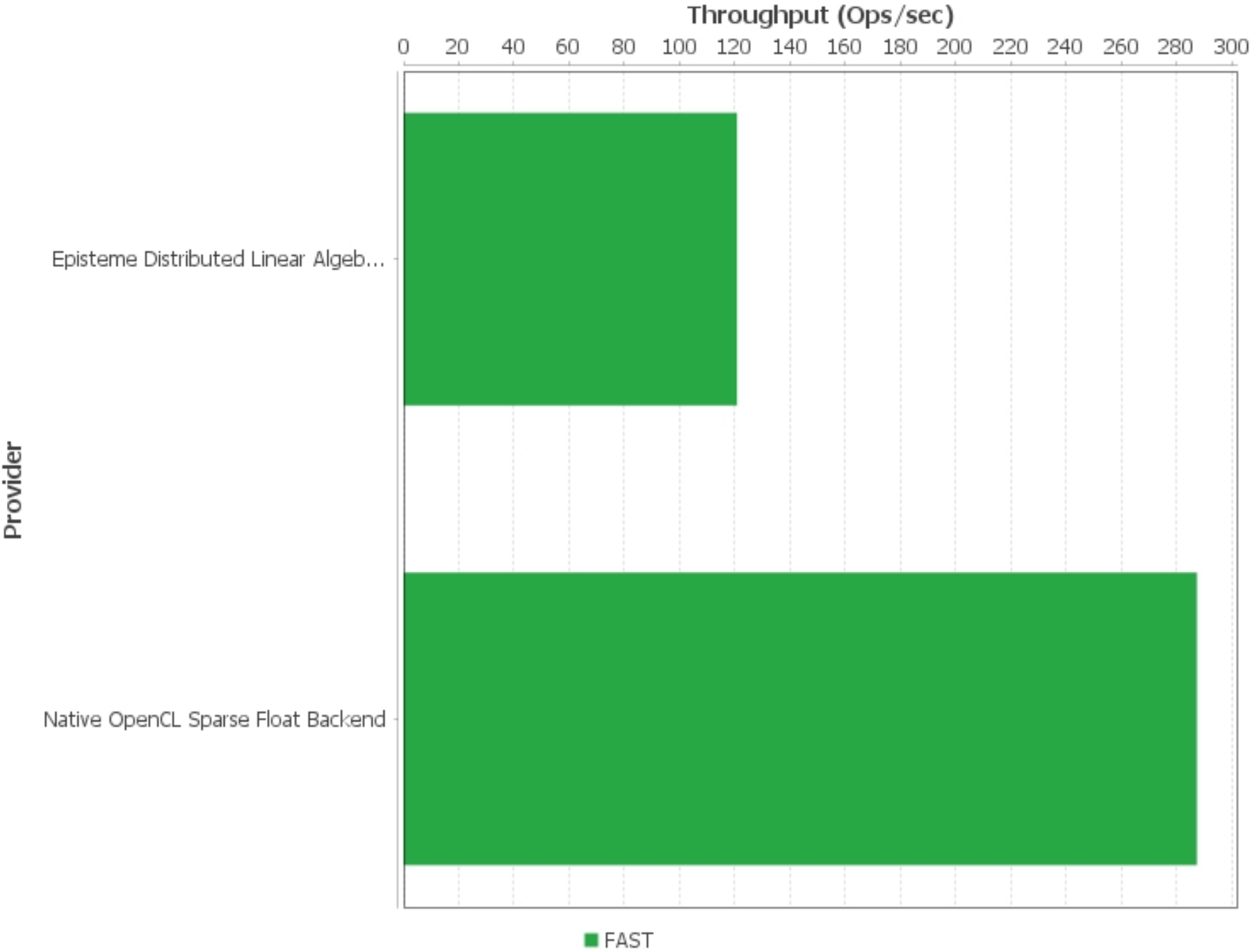
R:Sparse:BiCGSTAB Performance Comparison



Operation Comparison: R:Sparse:ConjGrad (FAST)

Episteme Multimodal Audit | R:Sparse:ConjGrad

R:Sparse:ConjGrad Performance Comparison



Operation Comparison: R:Sparse:GMRES (FAST)

Episteme Multimodal Audit | R:Sparse:GMRES

R:Sparse:GMRES Performance Comparison

